MIL-M-55649A(EL)

10 June 1977

Superseding
MIL-M-55649(EL)
24 December 1968

MILITARY SPECIFICATION
MULTIPLEXER TD-754( )/G

This specification is approved for use by Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for the manufacture and acceptance of the Multiplexer TD-754( )/G a 6-, 12-, 24- or 48 channel cable combiner. The unit provides power for the unattended Pulse Form Restorer, TD-206 and contains an integral order wire facility (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Material Handling, Wood, Stronger Construction 2 Way and 4 Way (Partial)
PPP-T-97	Tape, Pressure-sensitive Adhesive, Filament Reinforced
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-D-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box Fiberboard Corrugated Triple Wall
PPP-S-760	Strapping, Nonmetallic (and Connectors)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N. J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-M-13231	Marking of Electronic Items
MIL-P-116	Preservation-Packaging, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-I-46058	Insulation Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rubber or Plastic for Packaging Cable Assembly, Special Purpose, Electrical CX-11230( )/G
MIL-L-55583	Cable Assembly Adapter CX-10734( )/G
MIL-M-38510	Microcircuits, General Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definition of and Basic Requirements for Enclosures for Electrical and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-type Base
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-GL-323465	Gages for TD-754( )/G Multiplexer
SC-DL-528640	Extender Panel
SC-DL-525400	Multiplexer TD-754/G
SM-B-525535	Multiplexer TD-754/G Test Requirement

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SM-B-525530	Power Supply 12A1, 12A1A1, 12A1A2 Test Requirement
SM-B-525531	Order Wire 1, 12A2 Test Requirement
SM-B-525536	Order Wire 2 & Alarm Circuit, 12A3 Test Requirement
SM-B-525532	Transmit 12A4 Test Requirement
SM-B-525533	Cable Input 12A5 Test Requirement
SM-B-525534	Receive 12A6 Test Requirement
SC-DL-531012	Cable Assembly, Radio Frequency CG-1040( )/U (__FT __IN)
SC-DL-531016	Cable Assembly, Special Purpose CG-2437( )/TCC (__FT)
SC-DL-531017	Cable Assembly, Special Purpose CG-2438( )/TCC (__FT __IN)
SC-DL-531020	Cable Assembly, Special Purpose CX-7874( )/TCC
SC-DL-531021	Cable Assembly, Special Purpose CX-7875( )/TCC
SC-DL-525135	Cable Assembly, Power Electrical CX-11727( )/V (__FT __IN)
DL-SC-A-339551	Cable Assembly, Adapter CX-10734( )/G

PUBLICATIONS

DEPARTMENT OF THE ARMY

TM 11-5805-383-12	Operator and Organizational Maintenance Manual - Multiplexer TD-754( )/G

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawing and Data List SC-DL-525400, including the requirements for parts, materials and processes thereon.

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements herein, shall apply.

3.4 Signal Characteristics.

3.4.1 PCM Signal Characteristics (See 4.8.12). The pulsewidth of PCM signals, at the PCM OUT connector(when measured at the 50% amplitude points on the leading and lagging edges of the PCM signals) shall be 3.472 usec for 6 channels, 1.736 usec for 12 channels, 0.868 usec for 24 channels and 0.434 usec for 48 channels. The rise and fall times shall be 20 to 120 ns when measured from the 10% to

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90% amplitude points on the PCM signal. At the PCM OUT Connector, digital "1" shall be a level between +0.4 to -0.4 volts and digital "0" shall be a level between -1.6 and -2.4 volts.

3.4.2 TO CABLE Signal Characteristics. (See 4.8.14) The PCM signal at the TO CABLE connector shall have a pulsewidth of 180 to 230 nanoseconds (when measured at the 50% amplitude points of the leading and lagging edges of the TO CABLE PCM signal). At the TO CABLE Connector digital "1" shall be a level between 1.6 to 2.0 volts above the ac zero level (which is the digital "0" level). The PCM signal at the TO CABLE Connector shall have a jitter of less than 30 ns.

3.4.3 Timing Signal Characteristics. (See 4.8.12 and 4.8.30) Output signals at the TIM OUT-1 or TIM OUT-2 connector shall have a pulse peak 1.6 to 2.4 volts above a baseline of -1.8 to -2.3 volts. Ripple at the baseline shall be within ±0.5 volts. The rise and fall time shall be 20 to 120 nanoseconds (ns) when measured from the 10% to 90% amplitude points on the timing signal and the pulse-width shall be 80 to 150 ns when measured at the 50% amplitude points on the leading and lagging edges of the timing signal. The 50% amplitude point on the leading edge of the timing signal shall lead the 50% point of the PCM transitions by -45 ns to +80 ns. The timing pulses at the TIM OUT-1 or TIM OUT-2 connectors shall be either a 288 or a 576 Khz rate for six channel operation, 576 Khz rate for 12 and 24 channel operation, and a 576 or 2304 Khz rate for 48 AR (attended repeater) operation. The equipment shall meet specified performance requirements over a frequency range of ±.003% of the nominal input timing rate.

3.4.4 Quarter, Half, Three Quarter and One Mile Operation. The equipment shall recover the timing signal, and operate correctly when the first and last sections of a cable system are 1/4, 1/2, 3/4, and 1 mile lengths. The error rate shall not exceed one error per minute. (See 4.8.15)

3.4.4.1 Signal Detection. (See 4.8.24). The signal detection process shall operate over the input signal range of 20 to 50 millivolts peak-to-peak for 6/12 channel (576 Khz bit rate) and 48 channel (2304 Khz bit rate) operation. An orderwire signal of 15 volts, 2 Khz maximum frequency applied at the FROM CABLE connector shall not interfere with the detection process. Error free operation, defined as one error per minute or less, shall take place with a minimum ratio of peak signal to RMS noise of +20 dB for gaussian noise. For single frequency interference, error free detection shall take place with the peak signal to RMS interference ratios specified in Table I.

TABLE I

Carrier Frequency	Maximum Peak Signal To RMS Interference Ratio
300 kHz	-10 dB
500 kHz	+1 dB
800 kHz	+9 dB
1152 kHz	+15 dB
2304 kHz	+25 dB
3000 kHz	+24 dB
4500 kHz	+18 dB

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3.4.4.2 Automatic phasing (See 4.8.16). In the 24 channel mode, the equipment shall interleave the two 12 channel PCM signals on a bit by bit basis. In case of failure of either 12 channel PCM input signal, a dummy pulse train shall automatically be inserted in it's place. The two PCM signals will automatically be interleaved properly for any phase of PCM IN 2 signal.

3.4.4.3 Dummy Train Detection (See 4.8.29). Receipt of the dummy train shall cause the dummy train detector to produce a reset pulse. The reset pulse shall occur between 100 and 300 milliseconds after receipt of this signal, provided the signal is still present. Reset pulses shall occur only when the dummy train is present. The base of the reset pulse shall be 0 $\pm$.75V and the peak shall be +4.5 $\pm$.5V.

3.4.5 Cable Orderwire

3.4.5.1 Signalling and test tone (See 4.8.10, 4.8.22). The orderwire circuit oscillator shall operate at 1600 $\pm$32 hz for Signalling and 1100 $\pm$100 hz for a Test Tone over the operating temperature range as specified in 4.6.1. The signalling tone level shall be a minimum of 14 volts rms into 880 ohms and 310 millivolts rms into 50 ohms. When tested in accordance with 4.8.22, the 1600 $\pm$32Hz signal shall cause the Call Lamp to light and sound the buzzer. When tested in accordance with 4.8.30 and 4.8.31, orderwire communication shall be clear and intelligible.

3.4.5.2 Orderwire alignment. (See 4.8.10, 4.8.11, 4.8.20, 4.8.21) The TTL adjustment when properly aligned shall indicate in the green area of the TEST ALIGN meter and the level measured at test point J1 shall be 150 millivolts RMS. The Headset Amplifier output shall be 75 $\pm$15 millivolts when either the CRL or PRL are adjusted for hairline indication on the TEST ALIGN meter.

3.4.5.3 Transmit Orderwire Amplifiers. (See 4.8.17) When a 1 Khz, 150 millivolt RMS signal is applied to the Microphone Input, 160 to 310 millivolts RMS shall be present across the 50 ohm termination at the output of the Patch-Thru Transmit Amplifier and a minimum of 9 volts RMS shall be present across the 880 ohms termination at the output of the Cable Transmit Amplifier. The frequency response at 300 to 1700 hertz shall be within $\pm$1 db of the 1 khz reference level with the input/output levels and terminations cited above.

3.4.5.4 Receive Orderwire Amplifiers. (See 4.8.18 and 4.8.19) When a 1 khz, 45 millivolt RMS signal is applied across the Patch-Thru Connector, 50 to 100 millivolts RMS shall be present across the 150 ohm termination at the output of the Headset Output Amplifier and a minimum of 9 volts RMS shall be present across the 880 ohms termination at the output of the Cable Transmit Amplifier. With a 1 khz, 224 millivolt RMS signal applied to the Patch-Thru Connector, the output values above shall be obtained.

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3.4.5.5 Undesired Transmission Path Isolation. (See 4.8.17, 4.8.18, 4.8.19) Undesired signal level outputs shall be at least 26db below the desired signal level output for the Cable Transmit, Headset Amp Output and Patch-Thru Transmit amplifiers.

Signal Applied	Undesired Output	Desired Output
Microphone Input: 150 MV	Headset Amp Output (Note A)	Cable Transmit Amp: 9 Volts Min Patch-Thru Transmit Amp: 160 - 310 MV
Radio Receive Input: 45 MV, 224 MV	Patch-Thru Transmit Amp Output	Headset Amp Output: 50 - 100 MV Cable Transmit Amp: 9V (Note B)
Cable Receive Input 150 Mv, 16V	Cable Transmit Amp Output	Headset Amp Output: Adjustable to 75 MV with CRL Potentiometer (Note C) Patch-Thru Transmit Amp: 160 - 310 MV

Note A: Prior to test, Headset Amp Output adjusted to 75 MV with 150 MV to Cable Receive Input. Test performed with no Cable Receive Input.

Note B: PRL (Patch-Thru Receive Level) Control adjusted for specified output for each input level specified.

Note C: CRL (Cable Receive Level) Control adjusted for desired output for each input level specified.

3.4.6 Alarms

3.4.6.1 Traffic alarm. (See 4.8.7) The audible alarm shall sound and the TRAFFIC light illuminate upon loss of the PCM input signal. The BUZZER OFF switch shall silence the audible alarm. With the audible alarm silenced, re-application of the PCM input signal shall cause the audible alarm to sound and the TRAFFIC light to extinguish. Depressing the BUZZER OFF switch shall again silence the audible alarm without delay.

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3.4.6.2 Cable Current Alarm. (See 4.8.8) The audible alarm shall sound and the CABLE CUR light illuminate with no cable current at the TO CABLE connector. The BUZZER OFF switch shall silence the audible alarm. With the audible alarm silenced application of cable current shall cause the audible alarm to sound and the CABLE CUR light to extinguish. Depressing the BUZZER OFF switch shall again silence the audible alarm without delay.

3.4.7 Power Supply.

3.4.7.1 AC Input. (See 4.8.25) When tested in accordance with 4.8.25, the equipment shall not exhibit errors greater than 1 per minute and successfully provide orderwire communications and signalling when energized from 115 volts ac plus or minus 5%, 47.5 to 420 hz.

3.4.7.2 Output. (See 4.8.26) The +12, +5, and -6 volts dc power supply voltages shall be regulated to plus or minus 2%, with 10 millivolts RMS maximum ripple over the AC input ranges cited in 3.4.7.1.

3.4.8 Cable Power Supply.

3.4.8.1 Output. (See 4.8.27) Regulation at the 38 ma level shall be within ±3 percent with loading requirements from 0 to 600 volts and with the AC Input level and frequency ranges as cited in 3.4.7.1.

3.4.8.2 Automatic Shut-off. (See 4.8.28) The Cable Power Supply shall automatically shut off when the cable current decreases to 16 ±14 milliamperes or increases to 45 ±3 milliamperes. The equipment shall discharge the cable input voltage to a value of 30 volts or less within 1.0 second after the cable current is cut off or turned off.

3.4.8.3 Metering and Controls. (see 4.8.11) A hairline indication on the TEST ALIGN meter for the +12V, +5V and -6V SERVICE SELECT switch positions shall indicate power supply voltages of 12V ±3%, +5V ±3% and -6V ±3% respectively. The CABLE CUR METER SEL switch position shall provide a hairline indication on the TEST ALIGN meter for cable current of 38 ma ±3%. Yellow band indications on the TEST ALIGN meter shall indicate actual values of the signal being measured to within ±5%. Green band indications on the TEST ALIGN meter shall indicate presence of the signal being monitored. Absence of the signal being monitored shall cause an out of green band indication.

3.4.9 Cable Fault Location. (See 4.8.9, 4.8.23) With the cable system, consisting of Restorer, Pulse Form, TD-206( )/G, and Cable Assembly, Special Purpose, Electrical CX-4245( )/G or CX-11230( )/G operating, the cable fault location circuitry shall locate faulty restorers and cable short circuits to within ± one mile.

3.4.10 Digital Impedance Requirements. (See 4.8.13) The input impedance of the PCM-IN-1, PCM-IN-2 and TIM IN lines shall be 91 ohms ±10%.

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3.4.11 Service Condition Communication When tested in accordance with 4.8.30 and 4.8.31, voice communication over master and slave systems shall be clear and intelligible.

3.5 Preconditioning.

3.5.1 Electrical Preconditioning (Burn-in). (See 4.7.1) Prior to submission to the specified Quality Conformance Inspections, and subsequent to the bounce preconditioning of 3.5.2 each assembled Multiplexer TD-754( )/G consisting of chassis assembly, all panel assemblies, and the power supply, plus spare assemblies defined by the contract, shall be electrically preconditioned for a minimum of 48 hours. The last 24 hours of electrical preconditioning shall be failure free. The maximum preconditioning time allowable for any unit, assembly or subassembly to be delivered on contract shall be 168 hours. Each fabricated TD-745( )/G and spare assemblies shall be subjected to this process. All end equipments failing Group A inspection, may be repaired and tested, (to the end equipment Group A requirements) accepted and delivered without repeat of electrical pre-conditioning. Repair actions involving panel assemblies shall be accomplished through use of preconditioned panels.

3.5.2 Bounce Preconditioning. (See 4.7.2) Each TD-754( )/G assembly shall meet the requirements herein without subsequent processing after submission to the Bounce Preconditioning of 4.7.2. Each fabricated TD-754( )/G shall be subjected to this process.

3.6 Service Conditions. (See 4.6) The TD-754( )/G shall meet the Service Conditions listed below. Where a test is referenced, meeting the test shall be considered as compliance with the requirements:

3.6.1 Temperature. The following low and high temperature requirements shall be met:

3.6.1.1 Low. (See 4.6.1.1) The equipment shall be operated during and after testing as specified in 4.6.1.1, and shall meet the measurements of 4.8.30 at room ambient temperature and at -25°F. The equipment shall withstand exposure to a non-operating temperature of -65°F.

3.6.1.2 High. (See 4.6.1.2) During and after the testing specified in 4.6.1.2, the equipment shall operate and meet the requirements of 4.8.30 at room ambient and +125°F with simultaneous solar radiation of 360 BTU/sq. ft/hr. The equipment shall withstand exposure to a non-operating temperature of +154°F.

3.6.2 Moisture resistance (humidity). (See 4.6.2) During and after testing as specified in 4.6.2 the equipment shall meet the requirements for the measurements as specified in 4.8.30. The equipment shall exhibit no physical damage such as corrosion, rust, blistering swelling or deterioration of parts and materials.

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3.6.3 Low pressure (altitude) (See 4.6.3). During and after testing as specified in 4.6.3, the equipment shall be operable and meet the requirements of 4.8.31 at altitudes of 15,000 ft above sea level and shall withstand transportation at altitudes at 50,000 feet above sea level.

3.6.4 Vibration, Bounce and Shock.

3.6.4.1 Vibration, internal (See 4.6.4). During and after testing as specified in 4.6.4, the equipment shall meet full specification performance for the measurements specified in 4.8.31.

3.6.4.2 Bounce (vehicular) (see 4.6.5). The equipment shall meet the requirements for the tests specified in 4.8.31 after subjection to the test specified in 4.6.5.

3.6.4.3 Shock (bench handling). (See 4.6.6) The equipment in its servicing condition shall meet the requirements for the tests specified in 4.8.31 after subjection to the test specified in 4.6.6. There shall be no evidence of physical damage except minor surface abrasion.

3.6.5 Fungus (See 4.6.10). After the fungus test of 4.6.10, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, or sparse microbial growth (see 6.3) as evidenced by growth colonization, which includes branching and sporulation on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.3) shall not be included in this evaluation.

3.6.6 Dripproof (see 4.6.7). The equipment in normal operating condition shall meet the requirements of 4.8.31 when tested in accordance with 4.6.7 (with the rain cover removed and the connector caps removed). The equipment shall not leak or show evidence of leakage at the conclusion of the test.

3.6.7 Salt Fog (see 4.6.8). After the salt fog test of 4.6.8, the equipment, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface; or exfoliation (see 6.3) of metal. Electrical operation is not required.

3.6.8 Dust (Fine Sand) (see 4.6.9). After testing as specified in 4.6.9, the equipment shall meet the requirements for the tests specified in 4.8.31. Except for exterior paint damage on the case or front panel, the equipment shall exhibit no electrical or mechanical damage.

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3.7 Finish, Protective. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings (see 4.4).

3.8 Marking.

3.8.1 General. Marking shall conform to Specification MIL-M-13231. (See 4.4). Front panel marking shall be Group I as described in that specification.

3.8.2 Visibility. Wherever practicable, parts shall be mounted so that their identification markings will be readily visible with minimum disassembly of the equipment.

3.8.3 Serial Numbers (see 4.4 Marking). Each Multiplexer TD-754( )/G shall be serial numbered.

3.9 Microelectronic Integrated Circuits (see 4.4). All microelectronic integrated circuit devices used in this equipment shall meet the requirements of MIL-M-38510 for Class B devices and their individual drawing requirements.

3.10 Electromagnetic Interference Characteristics (see 4.9). The equipment shall meet the electromagnetic interference emission and susceptibility requirements of MIL-STD-461A, Notice 4, Table A-1, for Class CE equipment except that only tests CE04, CS02, CS06, RE02, RE02.1, and RS03 are applicable. Tests and test methods shall be as specified in MIL-STD-462, Notice 3.

3.11 Interchangeability (see 4.10). Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.12 Panel Assemblies (Electrical Test) (see 4.12). The TD-754( )/G and panel assemblies listed below shall meet the requirements of their respective test requirement drawings as specified in 4.12.

Multiplexer TD-754( )/G (TRS Test and Alignment Requirements) SM-B-525535

Power Supply	12A1	SM-B-525530	Panel Assy	12A3	SM-B-525536
Panel Assy	12A1A1	SM-B-525530	Panel Assy	12A4	SM-B-525532
Panel Assy	12A1A2	SM-B-525530	Panel Assy	12A5	SM-B-525533
Panel Assy	12A2	SM-B-525531	Panel Assy	12A6	SM-B-525534

3.13 Safety engineering.

3.13.1 Personnel hazards (see 4.11). Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

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3.13.2 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.14 Operational test (see 4.8.6). Each fabricated and tested Multiplexer TD-754( )/G shall be subjected to a Final Operational Test in accordance with 4.8.6 after all other inspections and tests are completed to determine that the TD-754( )/G is operational prior to packaging.

3.15 Workmanship (see 4.4 and 4.11). The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraph:

In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack.
- Protection.
- Clearance.
- Splicing.
- Connections, general.
- Grounding, general.
- Shielding on wire and cable.

Soldering and brazing.
Cleaning.
General requirements for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.

3.16 Printed wiring assemblies (see 4.4). Printed wiring assemblies shall be in accordance with the applicable paragraph in MIL-P-11268 and shall be conformally coated in accordance with MIL-STD-275 with a coating material that conforms to MIL-I-46058, type ER (Epoxy).

3.17 Reliability (see 4.13). The equipment shall demonstrate a specified (θ_o) mean-time-between-failure (MTBF) of 1436 hours, when tested in accordance with 4.13.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for Government Inspection shall be made available by the contractor.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3) - does not include preparation for delivery.

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.14).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article units. The contractor shall furnish 13 first article units of the complete Multiplexer TD-754( )/G.

4.3.2 First Article inspection. The first article inspection shall consist of inspections specified in Table II, and shall be performed in the order specified herein.

4.3.3 First Article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement in the contract or purchase order.

Item	Where Required
Parts, Materials, and Processes; General	3.3
Finish	3.7
Marking	3.8
Microelectronic Integrated Circuits	3.9
Workmanship	3.15
Printed Wiring Assemblies	3.16

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4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4, 4.7, and 4.5.1 through 4.5.5 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specified requirements.

4.5.1 Group A inspection. Each unit (and each subassembly and assembly identified in Table III on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. The order of inspection of the subassemblies and assemblies and units (end equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in Table III shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in Table III.

b. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table III (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies from the discrete lots which have passed the above Quality Assurance Verification Inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 3.5 and a 100% visual/mechanical and electrical inspection in accordance with Table III in an order satisfactory to the Government with the operational test performed last.

NOTE: Assemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's 100% Group A Inspection of the end equipment.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQL's indicated in Table III. Group A electrical inspection of the end equipment shall be performed in the order shown in Table III.

NOTE: Assemblies and subassemblies are not required to undergo further visual/mechanical and electrical inspection as part of the above contractor's sampling inspection of the end equipment.

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TABLE II - FIRST ARTICLE INSPECTION

Inspections	Rqt. Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6-13
1. Inspections covered by subsidiary documents		4.4	Inspection to be performed on all units					
2. Bounce preconditioning	3.5.2	4.7.2	Inspection to be performed on all units					
3. Electrical preconditioning	3.5.1	4.7.1	Inpsection to be performed on all units					
4. Group A Inspection		Table III	Inspection to be performed on all units					
5. Group B Inspection		Table IV	Inspection to be performed on all units					
6. Group C Inspection		Table V						
Low Temperature	3.6.1.1	4.6.1.1	3				1	
High Temperature	3.6.1.2	4.6.1.2	4				2	
Humidity	3.6.2	4.6.2					3	
Low Pressure (altitude)	3.6.3	4.6.3		1				
Vibration (internal)	3.6.4.1	4.6.4	1					
Bounce (vehicular)	3.6.4.2	4.6.5	2					
Shock (bench handling)	3.6.4.3	4.6.6				3		
Fungus (see Note d)	3.6.5	4.6.10			1			
Dripproof	3.6.6	4.6.7						
Salt Fog (see Note d)	3.6.7	4.6.8		3				
Dust	3.6.8	4.6.9		2				
EMI	3.10	4.9				1		
7. Group D Reliability	3.17	4.13						1 (Note c)

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NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all First Article units before subjecting these units to any other inspection requirements in the table.

b. The numbers in the unit columns in the table covered by Inspections numbered 6 and 7 specify the order of inspections for the indicated unit.

c. Units 6-13. These units are to undergo the same tests as units 1 through 5 up to and including Group B inspection. Thereafter, units 6 through 13 are to be subjected to Reliability Testing, para 4.13.

d. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

TABLE III GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL	
			Major	Minor
Visual-mechanical				
TD-754( )/G Case & Covers	3.15	4.11	2.5*	10.0*
Filter Assy 12A9	3.15	4.11	2.5*	4.0*
Filter Assy 12A8	3.15	4.11	2.5*	4.0*
Panel Assy 12A7	3.15	4.11	1.0*	4.0*
Panel Assy 12A6	3.15	4.11	1.0*	4.0*
Panel Assy 12A5	3.15	4.11	1.0*	4.0*
Panel Assy 12A4	3.15	4.11	1.0*	4.0*
Panel Assy 12A3	3.15	4.11	1.0*	4.0*
Panel Assy 12A2	3.15	4.11	1.0*	4.0*
Power Supply 12A1	3.15	4.11	1.0*	4.0*
Panel Assy 12A1-A1	3.15	4.11	1.0*	4.0*
Panel Assy 12A1-A2	3.15	4.11	1.0*	4.0*
Heat Sink 12A1-A3	3.15	4.11	2.5*	4.0*

*Defects per hundred units.

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TABLE III GROUP A INSPECTION (CONT'D)

Inspection	Req Para	Insp Para	Ref Figure	AQL Major	AQL Minor
Electrical Performance (*Including spare panel assy*)					
Power Supply 12A1	3.12	4.12		1.0%	**
Panel Assy 12A1-A1	3.12	4.12		1.0%	**
Panel Assy 12A1-A2	3.12	4.12		1.0%	**
Panel Assy 12A3	3.12	4.12		1.0%	**
Panel Assy 12A4	3.12	4.12		1.0%	**
Panel Assy 12A5	3.12	4.12		1.0%	**
Panel Assy 12A6	3.12	4.12		1.0%	**
Panel Assy 12A7	3.12	4.12		1.0%	**
Multiplexer TD-754/G	3.12	4.12		1.0%	**
Electrical					
Operational Test	3.14	4.8.6	2	1%	**
Traffic Alarm	3.4.6.1	4.8.7	3	1%	**
Cable Current Alarm	3.4.6.2	4.8.8	3	1%	**
Limited Fault Condition	3.4.9	4.8.9,4.8.23	3	1%	**
Signalling & Test Tone	3.4.5.1	4.8.10,4.8.22		1%	**
Orderwire Alignment	3.4.5.2	4.8.10,4.8.11, 4.8.20,4.8.21	4	1%	**
Metering & Controls	3.4.8.3	4.8.11	5	1%	**

** All electrical defects are major

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government with the Operational Test performed last.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table I of MIL-STD-105.

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TABLE IV GROUP B INSPECTION

Inspection	Req Para	Ref Fig	Insp Para	AQL
Interchangeability, Mechanical	3.11		4.10	6.5%
PCM & Timing Signal Characteristics	3.4.1, 3.4.3	6,7	4.8.12	4.0%
Digital Impedance	3.4.10	8	4.8.13	4.0%
TO Cable Signal Characteristics	3.4.2	9	4.8.14	4.0%
1/4, 1/2, 3/4, 1 Mile Operation	3.4.4	10	4.8.15	4.0%
Automatic Phasing	3.4.4.2	11	4.8.16	4.0%
Transmit Orderwire Amplifiers	3.4.5.3	12	4.8.17	4.0%
Receive Orderwire Amplifiers	3.4.5.4	13	4.8.18 4.8.19	4.0%
Undesired Transmission Path Isolation	3.4.5.5	12,13,14	4.8.17 4.8.18 4.8.19	4.0%
Orderwire Alignment	3.4.5.2	13,14	4.8.20 4.8.21	4.0%
Signalling and Test Tone	3.4.5.1	4,15	4.8.10, 4.8.22	4.0%
Cable Fault Location	3.4.9	16	4.8.23	4.0%
Signal Detection	3.4.4.1	17	4.8.24	4.0%
AC Input	3.4.7.1	18	4.8.25	4.0%
Power Supply Output	3.4.7.2	19	4.8.26	4.0%
Cable Power Supply Output	3.4.8.1,	20	4.8.27	4.0%
Metering and Controls	3.4.8.3	5	4.8.11	4.0%
Automatic Shut-Off	3.4.8.2	21,22	4.8.28	4.0%
Dummy Train Detection	3.4.4.3	23	4.8.29	4.0%

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4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in Table IV and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

TABLE V GROUP C INSPECTION

Inspection	Req Para	Ref Fig	Insp Para
Sub-Group 1			
Low Temperature	3.6.1.1	27	4.6.1.1, 4.8.30
High Temperature	3.6.1.2	27	4.6.1.2, 4.8.30
Moisture Resistance (Humidity)	3.6.2	27	4.6.2,4.8.30
Sub-Group 2			
Dripproof	3.6.6	27	4.6.7, 4.8.31
Vibration, internal	3.6.4.1	27	4.6.4, 4.8.31
Bounce, vehicular	3.6.4.2	27	4.6.5, 4.8.31
Shock (Bench Handling)	3.6.4.3	27	4.6.6, 4.8.31
EMI	3.10	33	4.9
Sub-Group 3			
Low Pressure (altitude)	3.6.3	27	4.6.3, 4.8.31
Dust (Fine Sand)	3.6.8	27	4.6.9 (See Note 1)
Salt Fog	3.6.7		4.6.8 (See Note 1)
Fungus	3.6.5		4.6.10 (See Note 1)

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried and refurbished if necessary, before proceding with subsequent tests.

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4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Sub-group 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every month. Each unit shall be subjected to Temperature and Humidity testing in that order.

4.5.3.1.2 Sub-group 2. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every third month. One unit shall be subjected to EMI, Dripproof, and Shock, (Bench Handling) testing; and one unit shall be subjected to Vibration (internal), and Bounce Vehicular testing with each unit subjected to testing in the order given.

4.5.3.1.3 Sub-group 3. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every fourth month. One unit shall be subjected to Altitude (Elevation), Dust and Salt Fog testing, in that order. One unit shall be subjected to Fungus testing.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

TABLE VI GROUP D INSPECTION

Inspection	Req Para	Insp Para
Reliability	3.17	4.13

4.5.5.1 Sampling for Group D inspection. Eight samples shall be selected at random from the first production lot. For subsequent Group D inspection, eight units shall be selected at random from each month's production lot.

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4.5.5.2 Group D failures. Actions required relative to Group D failures shall be in accordance with the procedure specified in MIL-STD-781.

4.5.5.3 Non-compliance. In case of Group D failure occurrence, procedures and requirements specified in the contract or purchase order shall be followed.

4.5.5.4 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Service condition tests. (see 3.6) The following service condition tests shall be performed. Since more than one unit is available, some tests may be run concurrently with Government approval by the contracting officer or his designated representative. After completion of the Fungus and Salt Fog tests, the equipment must be thoroughly washed, cleaned, dried and refurbished, if necessary before proceeding with subsequent tests.

4.6.1 Temperature Tests.

4.6.1.1 Low temperature. (See 3.6.1.1) The equipment shall be subjected to Test Method 502.1, Procedure I of MIL-STD-810. Storage temperature shall be -65 degrees F and operation temperature shall be -25 degrees F. Storage time shall be 24 hours. At operation temperature, tests shall start within 5 minutes after ac power application. A pretest performance record is required and is specified in 4.8.30. Performance checks per 4.8.30 shall be made as specified in Test Method 502.1 during tests.

4.6.1.2 High temperature (see 3.6.1.2). The equipment shall be subjected to Test Method 501.1, Procedure II of MIL-STD-810 except as follows: Step 4, the non-operating temperature is 68°C (154°F), Step 8 shall be modified as follows: Solar radiation of the intensity of 360 BTU/sq. ft/hr shall be maintained for four hours and then the equipment shall be tested in accordance with paragraph 4.8.30. The highest operating temperature is 125°F +5° -0°F not including the effects of solar radiation. Performance tests per 4.8.30 shall be made during tests as specified in Test Method 501.1, Procedure II.

4.6.2 Moisture Resistance test (humidity) (See 3.6.2). The equipment with the power supply, top, rear, and panel covers removed shall be subjected to Test Method 507.1, Procedure II of MIL-STD-810. Test measurements per 4.8.30 shall be taken during the 24 to 28 hr period and the last 5 hr period during each 48 hour cycle. Insert the Power Supply in the end item for electrical tests.

4.6.2.1 Failure. If any equipment fails to meet the requirements for the tests specified in 4.8.30 during cycling, it shall be realigned or readjusted once. If the equipment then fails or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.8.30 it does not pass the test.

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4.6.3 Low pressure (Altitude) (See 3.6.3). The equipment shall be subjected to the following test:

a. The equipment shall be placed in the test chamber. Prior to testing, the equipment shall meet the requirements for the tests specified in 4.8.31 except that auxiliary test equipment need not be inclosed in the chamber. A temperature of 73° ±18°F shall be maintained during the entire test.

b. Reduce the chamber internal pressure to 87.5 mm of Hg (3.44 in of Hg or 50,000 feet above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

c. Increase the chamber pressure to 429.1 mm of HG (16.9 in. of Hg or 15,000 feet above sea level) and then perform the test of 4.8.31 and record results.

d. With the equipment not operating, return the chamber to standard ambient conditions and perform the test of 4.8.31 and visually inspect the equipment.

e. During and after the test, the equipment shall meet the requirements for the test of 4.8.31 and shall not exhibit any electrical or mechanical degradation which could in any manner prevent the equipment from meeting any requirements of this specification.

4.6.4 Vibration test (internal) (see 3.6.4.1).

a. Resonant frequencies of the equipment shall be determined by varying the frequency of applied vibration slowly through the specified range at reduced test levels but with sufficient amplitude to excite the item.

b. The test item shall be attached to the vibration exciter table by its normal mounting means or by means of a rigid fixture capable of transmitting the vibration conditions. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test. The TD-754( )/G shall be mounted on the vibration table, positioned in its normal operating plane. The top and rear covers shall be removed. The card cell cover and connector covers shall be in place. The vibration shall be in a plane perpendicular to the 12A2 through 12A6 panels.

c. The test item shall be operated during the vibration resonance search so that functional effects caused by this test may be evaluated.

d. The test level shall be applied over a frequency range of 10 to 55 cps and at a vibratory displacement of not less than 0.030 inches, double amplitude. The vibratory frequency shall be changed in discrete steps of 1 cps and maintained at least 10 seconds at each frequency. Where possible, fixturing shall be such that reasonances within the test item can be observed and measured. The equipment shall have no resonances in the frequency range of 10 to 55 cps that exceed twice the amplitude of applied vibration.

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e. The equipment shall meet pretest and post test performance requirements in accordance with paragraph 4.8.31.

4.6.5 Bounce (vehicular) test (see 3.6.4.2). The equipment in its normal operating configuration with connector covers installed shall be subjected to Test Method 514.2, Procedure IX, Part 2 of MIL-STD-810. At the conclusion of test, the equipment shall meet full specified performance when tested in accordance with 4.8.31. Pretest and post test data per 4.8.31 is required.

4.6.6 Shock (bench handling) test (see 3.6.4.3). The equipment with connector covers installed shall be subjected to Test Method 516.2, Procedure V of MIL-STD-810. At the conclusion of test, the equipment shall meet full specified performance when tested in accordance with 4.8.31. Pretest and post test data per 4.8.31 is required. The Card Cell Cover shall be installed during all drops. All other covers shall be removed.

4.6.7 Dripproof test (see 3.6.6). The equipment shall be tested per MIL-STD-108, 4.3(b) Drip Test, except as follows:

a. The equipment shall be in its normal operating configuration.

b. In lieu of equipment orientation 15° from the vertical drops of liquid shall be falling at an angle of 15° from the vertical when the equipment is mounted in its normal position.

c. Immediately following the drip test, the equipment shall meet the full specified performance when tested in accordance with 4.8.31.

4.6.8 Salt Fog test (See 3.6.7). The equipment in its normal operating configuration with connector covers installed shall be subjected to the Salt Fog test specified in Method 509.1, Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. At the conclusion of the test, the equipment shall be examined for evidence of visible corrosion.

4.6.9 Dust test (see 3.6.8). The equipment, in its normal operating configuration, with connector covers installed shall be subjected to Test Method 510.1, Procedure I of MIL-STD-810. The equipment shall not be operated during the test. Pretest and post test data is required in accordance with 4.8.31.

4.6.10 Fungus test (see 3.6.5). The equipment in its normal operating configuration with connector covers installed shall be subjected to the fungus test specified in Method 508.1, Procedure 1 of MIL-STD-810. There shall be abundant growth colonization (6.3) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

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4.7 Preconditioning (see 3.5).

4.7.1 Electrical preconditioning inspection (burn-in) (se 3.5.1). Electrical preconditioning inspection shall meet the requirements of 3.5.1.

4.7.1.1 Test Details.

Electrical preconditioning (burn-in). Test level B of MIL-STD-781 shall apply for temperature and vibration requirements.

a. During testing, the equipment shall be mounted on a vibration table (in its normal operating configuration), and positioned in its normal operating plane with all covers in place. Vibration shall be in a plane perpendicular to the 12A2 through 12A6 printed circuit boards. During all operating periods, the equipment under test shall be electrically connected per Figure 1. The equipment shall be turned on and operated in the 48 AR channel mode for 23 hours and then turned off for 1 hour. The next 24 hour period, 23 hours operating and 1 hour off, shall be in the 12 channel mode as indicated in Fig 1. Time expended to perform the Operational tests shall be considered part of the "ON" cycle of the "ON-OFF" cycle. The Cable Power Supply shall be operational during the "ON" cycle. For these test 48 AR and 12 channel modes shall be used alterntately. At the end of each 24 hour period, the mode shall be changed. During each 24 hour period, the operational test per Table VIII shall be performed after an elapsed time of 12 + 2 hours. At the end of each operating period the test data shall be recorded.

TABLE VII

Test	Req Para	Fig	Test Para
Operational Test	3.14	2	4.8.6
Signaling and Test Tone	3.4.5.1	2	4.8.6
Signal Detection *(error rate)	3.4.4.1	2	4.8.6
Cable Current output	3.4.8.1	2	4.8.6
DC Power supply output	3.4.7.2	2	4.8.6

*NOTE measure error rate in either the 48 AR or 12 channel mode currently under test.

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b. The chassis, plug-in printed circuit panel assemblies and power supplies shall be considered to have been successfully electrically preconditioned when they have individually passed a cumulative 48 hour operating preconditioning period without a failure during the last 24 hour period. If during the last 24 hour operating preconditioning period a plug-in panel assembly, power supply, or chassis fails, it may be replaced with a preconditioned panel, power supply, or chassis and the preconditioning continued without the restart of a new 48 hour preconditioning period. A plug-in panel assembly, power supply or chassis failing operating preconditioning shall, after repair, be resubmitted to and pass the entire 48 hour operation preconditioning period prior to submission to subsequent quality conformance inspections.

c. The unit chassis shall have no less than 48 hours of operational preconditioning nor more than 168 hours of operational preconditioning (exclusive of production test time) when used as a vehicle for preconditioning power supplies and plug-in assemblies (including all spares).

d. Assemblies failing electrical acceptance tests subsequent to satisfactory completion of electrical preconditioning may be repaired with an electrically preconditioned assembly and electrically retested, accepted and delivered without repeat of electrical preconditioning.

4.7.1.2 Marking of electrically preconditioned units. All electrically preconditioned plug-in panel assemblies, power supplies, and chassis shall be physically identified by a preconditioning stamp. The stamp shall be affixed to the front panel near the nameplate of each unit which has successfully completed the preconditioning as outlined herein. The stamp on preconditioned plug-in panel assemblies shall be affixed to the bottom side and to the bottom outside edge of the Power Supply subassembly 12A1. The stamp shall be a "p" in a circle and shall be marked on the preconditioned unit per the Group III requirements of MIL-M-13231.

4.7.2 Bounce preconditioning (see 3.5.2). Each unit, in its non-operating condition, shall be placed in its normal operating position on the table of the package tester Type 1000 SC as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute prior to electrical preconditioning. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to electrical preconditioning.

4.8 Electrical tests.

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4.8.1 Standard conditions. Unless otherwise specified, all tests shall be performed at 115 volts ±5 percent from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperatures, pressure and humidity per MIL-STD-810 unless otherwise specified.

4.8.2 Alignment for normal operation.

a. Position Cable Mile switches on panels 12A4 and 12A5 as required.

b. Apply power to equipment.

c. Align power supply DC voltages to hairline (except +5 volt meter reference).

d. Adjust orderwire CRL (Cable Receive Level) control for green band indication.

e. Adjust orderwire PRL (Patch-Thru Receive Level) control for green band indication.

f. Turn on cable power.

g. Adjust cable current for hairline.

4.8.3 Standard test equipment. Standard test equipment is indicated in test figures by manufacturer and model number. In all cases, equivalent substitutes may be used. Government Technical Representative approval is required prior to use of proposed substitute items. The equipment indicated in the figures is listed in Table VII. Standard test equipment shall be provided by the contractor.

TABLE VIII STANDARD TEST EQUIPMENT

Designation in Figure	Function	Manufacture
Powertron 2505	AC Power Supply	Industrial Test Eqp Co. Inc.
Daven 12B	Transmission measuring set	Daven Div, Thomas A. Edison
GR1390B	Random noise generator	General Radio Co.
HP 201C	Audio Oscillator	Hewlett-Packard Co.
HP 200CD	Wide-range Oscillator	Hewlett-Packard Co.
HP 334A	Distortion Analyzer	Hewlett-Packard Co.
HP 400EL	Vacuum tube Voltmeter	Hewlett-Packard Co.
HP 5245L	Electronic counter	Hewlett-Packard Co.
HP 606B	Test Oscillator	Hewlett-Packard Co.
HP 6101A	DC Power Supply	Hewlett-Packard Co.
TEK R5403/D40	Dual Trace Oscilloscope	Tektronix Inc.
Clarostat 240C	Resistance Box	Clarostat Mfg. Co.
USVH	Frequency Selective Voltmeter	Rhode & Schwartz
Fluke 8120A	AC/DC Digital Voltmeter	Fluke, Inc.

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4.8.4 Special test equipment. Special test equipment is indicated by name in test figures. A description and block diagram of this equipment follows. Unless otherwise specified special test equipment shall be provided by the contractor subject to approval of the Government.

4.8.4.1 Medium traffic delay and buffer circuit. (see Figure 29) This circuit is to vary the phase of the PCM IN-2 signal applied to the Multiplexer TD-754( )/G with respect to the TIM IN signal.

4.8.4.2 Pseudo random pattern generator and error detector (PRPG). (see Figure 30) This is a circuit to provide a psuedo random PCM pattern at a 12 and 48 channel rate. The generator shall consist of a 23 or more stage shift register with the appropriate feed back circuitry to produce a maximal length sequence. An error detection circuit shall also be included for detection of transmission errors. For error detection, two generators are needed, one for transmit and one for receive. External timing of the correct frequency may be used. For waveform measurement, the number of shift register stages can be decreased. This shorter sequence shall be called the "short pattern" and the pattern generated by the 23 stage registers is the "long pattern". A single error generator shall be provided to verify proper operation of the PRPG prior to use.

4.8.4.3 Cable signal test circuit. (See Figure 31). This circuit is used to test the effect of noise and carrier on the Multiplexer TD-754( )/G. It allows for the addition of noise or tone to the PCM cable signal at the 1/2 mile point and at the same time allows adjustment of the level of the FROM CABLE PCM signal from 20 to 50 millivolts peak to peak at the next Multiplexer TD-754( )/G input.

4.8.4.4 Cable power supply test circuit. (See Figure 32). This circuit is used to provide various test loads for the Multiplexer TD-754( )/U cable power supply. It is also used to test the cable voltage transient.

4.8.4.5 Crystal controlled oscillator. This circuit shall consist of a crystal controlled oscillator whose output frequency can be varied from 575982 Hz to 576018 Hz. This circuit shall be capable of slaving a Multiplexer TD-660( )/G through the SYNC IN connection.

4.8.4.6 Dummy train test circuit. (See Figure 23) This circuit will detect the dummy train reset pulses between 100 and 300 milliseconds after receipt of the signal.

4.8.5 Ancilliary items. Ancilliary items required for testing by this specification shall be furnished by the contractor in quantities necessary and with cable lengths as specified herein.

4.8.5.1 Cable assembly, power, electrical CX-11727( )/U. In accordance with SC-DL-525135, length shall be five (5) feet unless otherwise specified in the bid request and contract.

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4.8.5.2 Cable assembly, radio frequency CG-1040( )/U. In accordance with SC-DL-531012, length shall be six (6) feet unless otherwise specified in the bid request and contract.

4.8.5.3 Cable assembly, radio frequency CG-2437( )/TCC. In accordance with SC-DL-531016, length shall be ten (10) feet unless otherwise specified in the bid request and contract.

4.8.5.4 Cable assembly, radio frequency CG-2438( )/TCC. In accordance with SC-DL-531017, length shall be ten (10) feet unless otherwise specified in bid request and contract.

4.8.5.5 Cable assembly, special purpose, electrical CX-7874( )/TCC. In accordance with SC-DL-531020, length shall be five (5) feet unless otherwise specified in the bid request and contract.

4.8.5.6 Cable assembly, speical purpose, electrical CX-7875( )/TCC. In accordance with SC-DL-531021, length shall be twenty-two (22) feet unless otherwise specified in the bid request and contract.

4.8.5.7 Extender panel. The extender panel provides for operation of one panel (12A2 through 12A6) with all components, controls, connections and test points accessible for troubleshooting. The extender panel shall be in accordance with SC-DL-528640.

4.8.5.8 Extender cable (12A1 Power Supply). The extender cable shall consist of a multiconduct cable with a 24 pin connector at each end. It shall be constructed for insertion between the chassis and the plug-in power supply sub-assembly and used for maintenance and alignment. This cable shall not be less than 18 inches in length or more than 20 inches in length.

4.8.5.9 Cable assembly Adapter CX-10734( )/G. Assembly shall be constructed in accordance with and meet the requirements of DL-SC-A-339551 and MIL-C-55583(EL).

4.8.6 Operational test.

a. Connect equipment per Figure 2 with cable power "ON".

b. Align equipment for 48 channel operation.

c. Measure error rate of equipment.

d. Align equipment for 12 channel operation.

e. Measure error rate.

f. Connect Pseudo Random Pattern Generator and Error Detector to the PCM OUT-2 TIM OUT-2 connector.

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g. Measure error rate.

h. Talk, signal and listen on cable orderwire using Handset H-156( )/U.

i. Measure cable power supply output current of the unit under test.

j. Measure DC power supply voltages of the unit under test.

k. Measure the frequency of the signalling oscillator and test tone oscillator of the unit under test.

l. The requirements of 3.14, 3.4.4.1, 3.4.5.1, 3.4.7.2 and 3.4.8.1 shall be met.

4.8.7 Traffic alarm test.

a. Connect equipment per Figure 3.

b. Align equipment for 12 channel operation.

c. Using capacitor and switch S1, short out cable signal at half mile point.

d. Note that the TRAFFIC lamp lights and the audible alarm sounds.

e. Silence audible alarm using pushbutton switch.

f. Disconnect switch S1.

g. Note that the TRAFFIC lamp is extinguished and that the audible alarm sounds.

h. Silence audible alarm using pushbutton switch.

i. The requirements of 3.4.6.1 shall be met.

4.8.8 Cable current alarm test.

a. Connect equipment per Figure 3.

b. Align equipment for 12 channel operation.

c. Operate equipment with cable power switch in OFF position. Note that the CABLE CUR lamp lights and the audible alarm sounds.

d. Silence audible alarm using pushbutton switch.

e. Turn cable power ON and note that the CABLE CUR lamp is extinguished and the audible alarm sounds.

f. Silence audible alarm using pushbutton switch.

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g. Open cable at switch S2. (Use caution as HIGH VOLTAGE may be present on line until current is switched off). Note that the CABLE CUR lamp lights and the audible alarm sounds.

h. The requirements of 3.4.6.2 shall be met.

4.8.9 Limited fault location test.

a. Connect equipment per Figure 3.

b. Align equipment for 12 channel operation.

c. Determine number of good Restorers TD-206( )/U in test system between TD-754( )/G units. Facility shall indicate there are two good restorers in the system.

d. The requirements of 3.4.9 shall be met.

4.8.10 Test Tone test.

a. Connect equipment per Figure 4.

b. Align equipment for 48 channel operation.

c. Set TONE SWITCH to ON position.

d. Measure TEST TONE LEVEL at J1 on panel 12A2.

e. Measure frequency of TEST TONE at TO CABLE connector.

f. The requirements of 3.4.5.1 and 3.4.5.2 shall be met.

4.8.11 Metering test.

a. Connect equipment per Figure 5.

b. Align equipment for 24 channel operation.

c. Using service facility controls, determine proper operation of all metering circuits by performing each of the following:

(1) Go-no-go indications (green area). Disable signal being monitored and note meter indication and note that the meter indicates a disabled condition.

(2) Critical adjustments (hairline). Adjust signals for hairline indication and note actual levels being monitored.

(3) 5% tolerance signals (yellow area). Measure actual values of signals being monitored.

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(4) Check power supply voltages.

d. The requirements of 3.4.8.3 and 3.4.5.2 shall be met.

4.8.12 Digital signal test.

a. Connect equipment per Figure 6.

b. Align equipment for 12 channel operation.

c. Measure characteristics of PCM and timing signals.

d. Measure phasing of PCM and timing signals.

e. Align equipment for 24 channel operation.

f. Repeat c and d above.

g. Connect equipment per Figure 7.

h. Align equipment for 48 channel operation.

i. Repeat c and d above.

j. The requirements of 3.4.1 and 3.4.3 shall be met.

4.8.13 Digital impedance test.

a. Connect equipment per Figure 8.

b. Adjust DC power supply for -1.0 dc at TD-754( )/G input.

c. Adjust oscillator for 10.9 volts rms at 10 Khz output.

d. Measure input impedance of PCM IN-1, PCM IN-2 and TIM IN connector.

e. The requirements of 3.4.10 shall be met.

4.8.14 Cable signal tests.

a. Connect equipment per Figure 9.

b. Align equipment for 6 channel operation.

c. Measure characteristics of TO CABLE signal. Pulse height and width shall be measured on first digital "1" following 6 consecutive digital "0"'s.

d. The requirements of 3.4.2 shall be met.

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4.8.15 1/4, 1/2, 3/4 and 1 mile test.

a. Connect equipment per Figure 10.

b. Align equipment for 48 channel operation.

c. Measure errors for 1 minute.

d. Remove 1/4 mile of Cable Assembly, Special Purpose Electrical CX-4245( )/G or CX-11230( )/G between Multiplexer TD-754( )/G and Restorer Pulse Form TD-206( )/G and adjust Cable Mile switches on panels 12A4 and 12A5.

e. Measure errors for 1 minute.

f. Repeat d and e above for 1/2 mile cable between TD-754( )/G and TD-206( )/G.

g. Repeat d and e above for 1/4 mile cable between TD-754( )/G and TD-206( )/G.

h. The requirements of 3.4.4 shall be met.

4.8.16 Automatic phasing test.

a. Connect equipment per Figure 11.

b. Align equipment for 24 channel operation.

c. Apply internal test tone to channel 1 of both TD-660's.

d. Monitor audio output of channel 1 of both TD-660's while varying the phasing of the PCM IN-2 signal.

e. The requirements of 3.4.4.2 shall be met.

4.8.17 Transmit amplifiers test.

a. Connect equipment per Figure 12.

b. Align equipment for 48 channel operation.

c. Adjust signal generator for 150 millivolts RMS at 1,000 hz and apply to FROM CABLE input.

d. Adjust CRL control for 75 MV Headset Output Amp level.

e. Remove signal from FROM CABLE input.

f. Adjust signal generator for 150 millivolts RMS at 1,000 hz and apply to Microphone input.

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g. Measure output of Patch-Thru Transmit, Cable Transmit and Headset Output Amplifiers.

h. Repeat f and g above for 300 hz and 1700 hz test tones.

i. The requirements of 3.4.5.3 and 3.4.5.5 shall be met.

4.8.18 Receive Orderwire Amplifier Test(Radio/Patch-Thru)

a. Connect equipment per Figure 13.

b. Align equipment for 48 channel operation.

c. Adjust signal generator for 45 millivolts RMS at 1,000 hz and apply to PATCH THRU connector.

d. Adjust PRL Potentiometer for 9 volts output from the CABLE TRANSMIT Amp.

e. Measure output of Patch Thru Transmit, Cable Transmit and Headset Output Amplifier.

f. Repeat c and e above for 300 hz and 1700 hz.

g. Adjust signal generator for 224 millivolts RMS at 1,000 hz and apply to PATCH THRU connector.

h. Repeat d above.

i. Measure output of Patch-Thru Transmit, Cable Transmit and Headset Output Amplifiers.

j. Repeat g and i above for 300 hz and 1700 hz.

k. The requirements of 3.4.5.4 and 3.4.5.5 shall be met.

4.8.19 Receive Orderwire Amplifier Test (cable rec).

a. Connect equipment per Figure 14.

b. Align equipment for 48 channel operation.

c. Adjust signal generator to 150 millivolts RMS at 1,000 hz and apply to FROM CABLE input.

d. Adjust CRL control for 75 mv Headset Output Amplifier level.

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e. Measure outputs of Patch Thru Transmit Amplifier, Headset Output Amplifier and Cable Transmit Amplifier.

f. Repeat c and e above for 300 hz and 1700 hz.

g. Repeat c, d, e and f above for 16 volts RMS generator output.

h. The requirements of 3.4.5.4 and 3.4.5.5 shall be met.

4.8.20 Cable Receive Level Alignment Test.

a. Connect equipment per Figure 14.

b. Align equipment for 48 channel operation.

c. Apply a 1V RMS - 1000 hz signal to FROM CABLE input.

d. Using SERV FAC adjust CRL control for hairline indication on TEST ALIGN METER.

e. Measure output of Headset Output Amplifier.

f. The requirements of 3.4.5.2 shall be met.

4.8.21 Patch-Thru Receive Level Alignment Test.

a. Connect equipment per Figure 13.

b. Align equipment for 48 channel operation.

c. Apply a 224 millivolt RMS signal at 1,000 hz to the Patch Thru connector.

d. Using the service facility adjust PRL for a hairline indication on the TEST ALIGN meter.

e. Measure output of Headset Output Amplifier.

f. The requirements of 3.4.5.2 shall be met.

4.8.22 Signalling test.

a. Connect equipment per Figure 15.

b. Align equipment for 48 channel operation.

c. Press TALK SIGNAL switch to SIGNAL position.

d. Measure level of Tone at TO CABLE connector and PATCH THRU connector.

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e. Measure frequency of signaling oscillator.

f. Place TALK SIGNAL switch to TALK position.

g. Apply a 224 millivolt RMS 1,100 hz signal to the PATCH THRU connector and adjust PRL for hairline indication.

h. Apply a 224 millivolt RMS 1,600 hz signal to the PATCH THRU connector and note operation of CALL LAMP and AUDIBLE SIGNAL.

i. Repeat h above at 1632 hz and 1568 hz.

j. Apply a 1 volt RMS 1100 hz signal to the FROM CABLE input and adjust CRL for hairline indication.

k. Apply a 1 volt RMS 1600 hz signal to the FROM CABLE input and note operation of CALL LAMP and AUDIBLE SIGNAL.

l. The requirements of 3.4.5.1 and 3.4.6.2 shall be met.

4.8.23 Fault location test.

a. Connect equipment per Figure 16.

b. Align equipment for 12 channel operation.

c. Simulate a fault at the input of the first Restorer, Pulse Form TD-206( )/G.

d. Using fault locating facilities, locate position of fault.

e. Repeat c and d above for simulated faults at 4th, 10th, 21st, 29th, 31st and 39th Restorer, Pulse Form TD-206( )/G.

f. The requirements of 3.4.9 shall be met.

4.8.24 Signal detection test.

a. Connect equipment per Figure 17.

b. Align equipment for 12 channel operation per paragraph 4.8.2.

c. Adjust level of FROM CABLE signal to 50 millivolts peak-to-peak (8.84 millivolts RMS at 576 khz bit rate for 50% "1" pattern) with switch S1 in position "1".

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d. Insert gaussian noise at half mile point and adjust for peak signal to RMS noise ratio of 20 db. Return switch S1 to position 2.

e. Apply orderwire signal of 15V at 2 khz to the FROM CABLE connector.

f. Press TALK - SIGNAL switch to SIGNAL position.

g. Count errors for 2 minutes.

h. Repeat c through g above for 30 mv and 20 mv peak to peak FROM CABLE signal. (5.3 and 3.5 millivolts RMS at 576 khz for a 50% "1" pattern).

i. Adjust level of FROM CABLE signal to 50 mv peak to peak, with Switch S1 in position 1.

j. Disconnect noise generator and connect oscillator.

k. Insert 300 khz sine-wave signal at input to "FROM CABLE" and adjust level for peak signal to RMS interference ratio of -10 db. Return switch S1 to position 2.

l. Count errors for 2 minutes.

m. Repeat i, j. and k above at 500 khz, 800 khz, 1152 khz, 2304 khz, 3000 khz and 4500 khz using values of peak signal to RMS interference ratio shown in Table I. (see 3.4.4.1)

n. Repeat i through m above for 30 mv and 20 mv peak to peak.

o. Repeat c through n above with the equipment aligned for 48 channel operation per paragraph 4.8.2.

p. The requirements of 3.4.4.1 shall be met.

4.8.25 AC input test.

a. Connect equipment per Figure 18.

b. Adjust variable power supply for 60 hz and 115 volts.

c. Align equipment for 12 channel operation.

d. Count errors for 2 minutes.

e. Talk, signal, and listen on cable orderwire.

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f. Repeat d and e for 47.5 hz 109 volts; 47.5 hz 121 volts; 400 hz 115 volts; 420 hz 109 volts; and 420 hz 121 volts.

g. The requirements of 3.4.4.1, 3.4.5.1 and 3.4.7.1 shall be met.

4.8.26 Power supply output test.

a. Connect equipment per Figure 19.

b. Adjust variable power supply for 60 hz and 115 volts.

c. Align equipment for 12 channel operation.

d. Measure DC power supply output voltages and ripple.

e. Repeat d for 47.5 hz 109 volts; 47.5 hz 121 volts; 400 hz 115 volts; 420 hz 109 volts; and 420 hz 121 volts.

f. The requirements of 3.4.7.2 shall be met.

4.8.27 Cable power supply test.

a. Connect the equipment per Figure 20.

b. Align the equipment for 12 channel operation.

c. With AC input set to 60 hz and 115 volts and load set for 7.5 k ohms adjust CABLE CURRENT adjust for hairline indication on TEST ALIGN meter.

d. Measure cable current.

e. Determine range of adjustment of cable current output.

f. With AC input set to 60 hz and 115 volts and load set for 7.5K ohms, adjust CABLE CURRENT for 38 milliamperes output.

g. Measure cable current for short circuit, 1K, 2K, 7.5K and 15K ohms loads.

h. Repeat g with AC input adjusted for 47.5 hz 109 volts; 47.5 hz 121 volts; 400 hz 115 volts; 420 hz 121 volts.

i. The requirements of 3.4.8.1, and 3.4.8.3 shall be met.

4.8.28 Automatic shut-off test.

a. Connect equipment per Figure 21.

b. Align equipment for 12 channel operation.

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c. Set cable load to 7.5K ohms.

d. Increase cable current until Over Current relay shuts off supply and note current.

e. After automatic shut off, the cable power switch shall be turned to the OFF position and back to the ON position before the cable current can be turned ON.

f. Adjust cable current for hairline indication.

g. Increase cable load resistance until Under Current relay shuts off supply and note current.

h. Repeat e.

i. Connect equipment per Figure 22.

j. Align equipment for 12 channel operation.

k. Open circuit cable current at far end of system while monitoring output voltage and note the time for the output voltage to decrease to 30 volts or less.

l. The requirements of 3.4.8.2 shall be met.

4.8.29 Dummy Train Detection test.

a. Connect equipment per Figure 23.

b. Align for 12 channel operation.

c. Adjust pulse generator output to 300 milliseconds.

d. Measure delay of reset pulse after application of dummy train.

e. Adjust pulse generator output to 100 milliseconds.

f. Note presence or absence of reset pulse.

g. The requirements of 3.4.4.3 shall be met.

4.8.30 Service condition test (Temperature, Humidity).

a. Connect equipment per Figure 27.

b. Align equipment for 24 channel operation. (This step is performed only once during entire temperature or humidity test cycle).

c. Talk and listen on channels 1 through 12, in both the master and slave modes of operation.

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d. Measure DC power supply voltages of unit under test.

e. Measure frequency of signaling oscillator and test tone oscillator in unit under test.

f. Measure Cable Power Supply cable current of the unit under test.

g. Apply external 1600 hz signal to TD-754( )/G "A" and note proper operation of signaling detector in unit under test.

h. Repeat g above with external frequency of 1568 hz and 1632 hz.

i. Apply external 576,018 Hz signal to SYNC IN master TD-660( )/G.

j. Repeat c above.

k. Repeat i and j with frequency set at 575,982 hz.

l. Repeat i, j and k when applying external signal to master TD-660( )/G at other end of system.

m. Signal, talk and listen on system orderwire in both directions.

n. The requirements of 3.4.7.2, 3.4.5.1, 3.4.8.1, 3.4.3 and 3.4.11 shall be met.

4.8.31 Service condition test (Dripproof, Vibration, Bounce, Shock, Low Pressure and Dust).

a. Connect equipment per Figure 27.

b. Align equipment for 24 channel operation.

c. Talk and listen on channels 1 through 12, in both the master and slave modes of operation.

d. Measure power supply DC voltages of unit under test.

e. Measure Cable Power Supply output current of unit under test.

*f. Signal talk and listen on system orderwire in both directions.

g. The requirements of 3.4.11, 3.4.7.2, 3.4.8.1, and 3.4.5.1 shall be met.

* DO NOT SIGNAL AT THE HIGHEST OPERATING ALTITUDE IN THE ELEVATION TEST.

4.9 Electromagnetic compatibility. (See 3.10) The equipment shall be tested for compliance to the requirements of MIL-STD-461, Notice 4, Table A-1, for Class CE equipment. The test methods and procedures described in MIL-STD-462, Notice 3 shall be employed. All electromagnetic compatibility tests shall be performed

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with the equipment simulating normal operation. The frequency range of the following tests are modified as follows: RE02 -14 khz to 1000 Mhz and RS03 -14 Khz to 1000 Mhz.

a. Connect equipment per Figure 33.

b. Align equipment for 12 channel operation.

c. Perform radiated and conducted emission tests on the equipment in accordance with MIL-STD-462 Notice 3 and the approved EMC/EMI test plan.

d. Perform radiated and conducted susceptibility tests on the equipment in accordance with MIL-STD-462 Notice 3 and the approved EMC/EMI test plan while monitoring the following:

(1) Listen on one audio channel.

(2) Listen on cable orderwire.

e. The requirements of 3.10 shall be met.

4.10 Inspection for dimensional interchangeability. The gage points identified on SC-GL-323465 shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11.

4.11 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252 for conformance with MIL-STD-454 and the requirements of 3.3 and 3.15. An inspection shall be performed to verify compliance with those portions of 3.13 (Safety Engineering) which can be determined visually.

4.12 Electrical subassembly test. (See 3.12) Multiplexer TD-754( )/G and plug-in subassemblies, including running spare panel assemblies, shall be tested in accordance with their respective test requirement specification drawings listed in 3.12.

4.13 Reliability. (See 3.17)

4.13.1 Reliability testing.

4.13.1.1 First Article reliability test. The First Article reliability inspection testing shall be in accordance with Test Plan IV of MIL-STD-781.

4.13.1.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan IV A of MIL-STD-781.

4.13.1.3 Reliability test conditions. The detail test conditions, procedures and methods for the test described in 4.13.1 and 4.13.2 above shall be in accordance with the approved Reliability Test Plan for Multiplexer TD-754( )/G specified elsewhere in the contract.

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4.13.2 Reliability test procedure.

4.13.2.1 Performance parameters to be measured. The tests specified in 4.8.6 shall be performed on all units subjected to Reliability testing.

4.13.2.2 Environmental and operational stresses.

4.13.2.2.1 Test level B of MIL-STD-781 shall apply except as noted below:

a. Input Voltage Cycling specified in Test level B does not apply.

b. Equipment ON-OFF cycling per Test level B is revised to read "Equipment ON-OFF Cycling". The equipment shall be turned on and operated in the 12 channel mode for 23 hours and then turned off for one hour. The next 24 hour period shall be in the 48 AR channel mode as indicated in Figure 1 with 23 hours operating and 1 hour off. Time expended to perform the operational tests shall be considered part of the "ON" period of the "ON-OFF" cycle. For these tests, 48 AR channel and 12 channel modes shall be used alternately. At the end of each 24 hour operating period the Operational Test (48 AR or 12 channel mode) of 4.8.6 (Table VI) shall be performed during the twenty-third hour and the test data shall be recorded. The test measurements shall not extend beyond the twenty-third hour.

4.13.2.2.2 Vibration. During testing, the equipment shall be mounted on a vibration table, positioned in its normal operating plane with all covers in place. Vibration shall be in a plane perpendicular to 12A2 through 12A6 printed circuit boards. Covers may be removed as necessary during test periods, but must be replaced at the conclusion of the electrical test period.

4.13.2.3 Maintenance measurements. Daily operator adjustments and preventive maintenance shall only be made during the one hour "OFF" period. The extent of the adjustments and preventive maintenance permitted is only that outlined for the operator in the organizational Technical Manual (TM) 11-5805-383-12. These adjustments shall not be considered failures.

4.13.2.4 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.13.2.5 Accept-reject criteria. The accept-reject criteria shall be either Test Plan IV or Test Plan IVA of MIL-STD-781 depending on the phase of testing being conducted.

4.14 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.(2)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Multiplexer TD-754( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each MUltiplexer TD-754( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be individually unit packed Method 1C-1.

5.1.1.4.2 Multiplexer TD-754( )/G. Each multiplexer shall be individually unit packed method III as follows: Cushion the item on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned item shall be inclosed within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specifed in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Items shall be given the best degree of protection employed by the supplier to affort protection against corrosion, deterioration, and damage during shipment. Items weighing over 10 pounds shall be individually packed.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.(2)b). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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5.2.1 Level A. A quantity of Multiplexer TD-754( )/G, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-P-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per desintation comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Multiplexer TD-754( )/G, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Multiplexer TD-754( )/G, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

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5.2.3 Commercial packing. The item shall be packed to meet carrier acceptance and safe delivery to destination at lowest rates in compliance with requirements of carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall as a minimum be marked as follows: Noun nomenclature; National Stock Number; or part number when NSN is not given; Government contract or purchase order number; quantity; contractor's name; additional markings as may be required by the contractor's policy/procedures; exterior containers shall also be marked with the appropriate address markings. Markings may be applied by any means which provide legibility.

6. NOTES

6.1 Intended use. Multiplexer TD-754( )/G operates in ATACS PCM assemblages having cable transmission capabilities.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B or commercial preservation and packing (see section 5).

c. When rough handling and functional test are required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required, test tools and running spares.

g. First Article Inspection (See 3.1).

h. Length of cables (if any are required).

6.3 Definitions.

6.3.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

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6.3.2 Exfoliation. Exfoliation is corrosion along the grain boundries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose pages or onion skin peeling.

6.3.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.4 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.3.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.4 Group C inspection. (See paragraph 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. (See paragraph 4.5.3) Initial shipments of end items will be contingent only upon successful completion of all Group C tests.

6.5 Location of operational inspection. (See paragraph 4.8.6) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TD-754W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

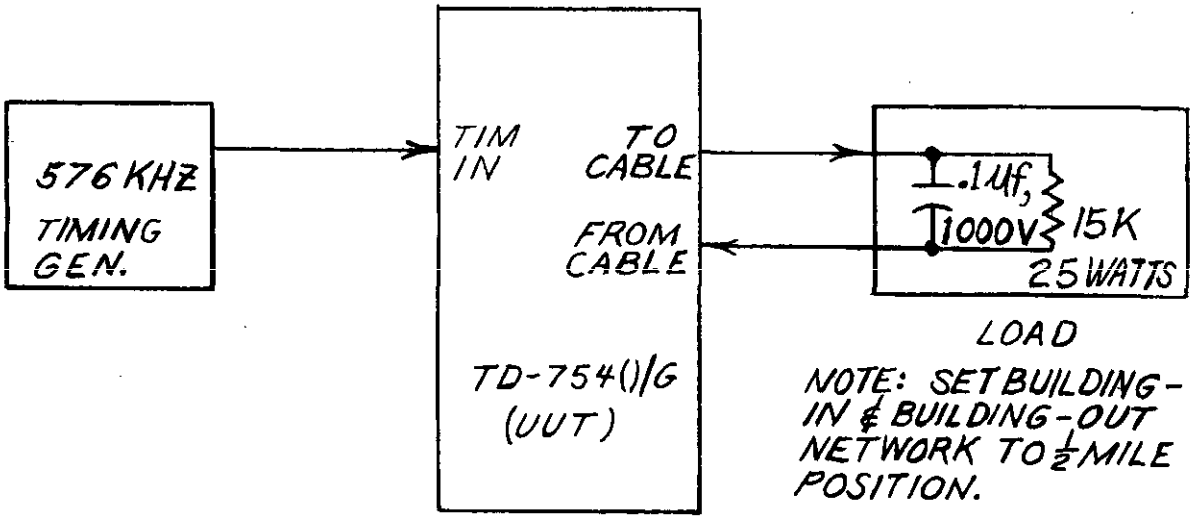
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6.8 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

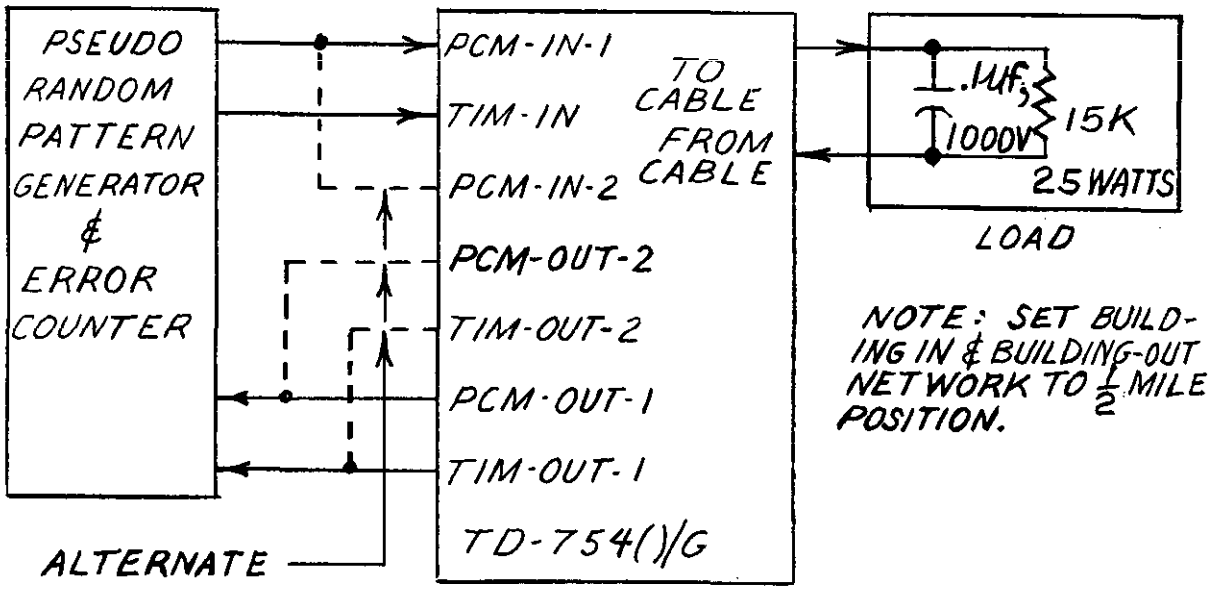
CUSTODIAN
ARMY-EL

Preparing Activity
ARMY-EL
Project No. 5805-A242

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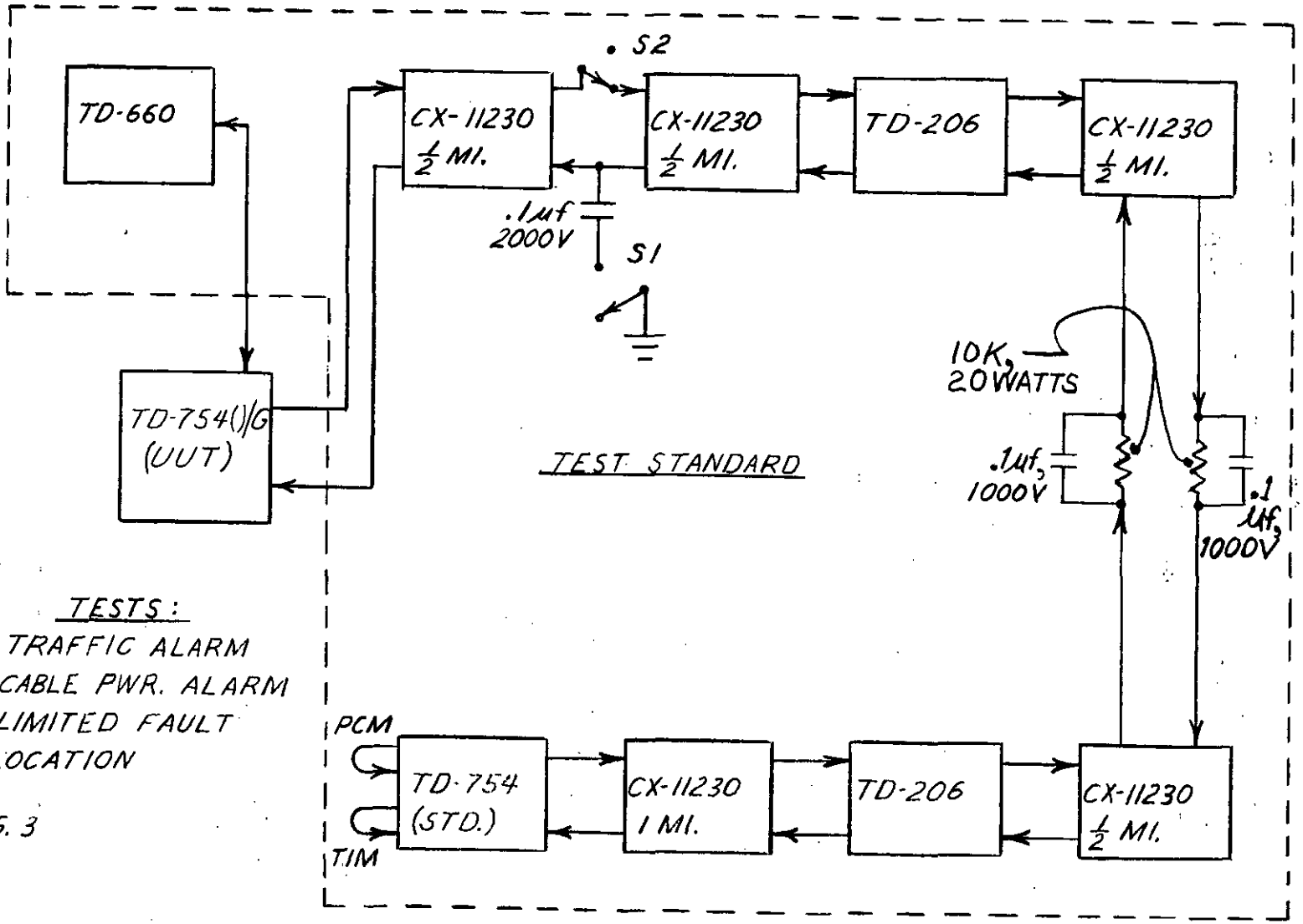


ELECTRICAL PRECONDITIONING (BURN-IN)
FIG. 1



OPERATIONAL TEST
FIG. 2

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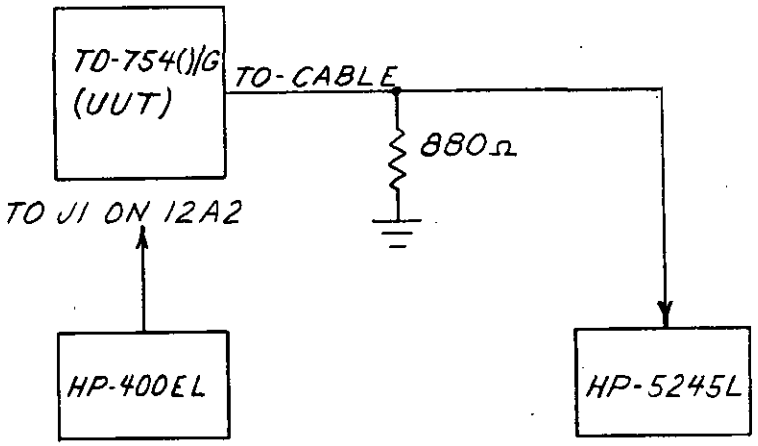


TESTS:

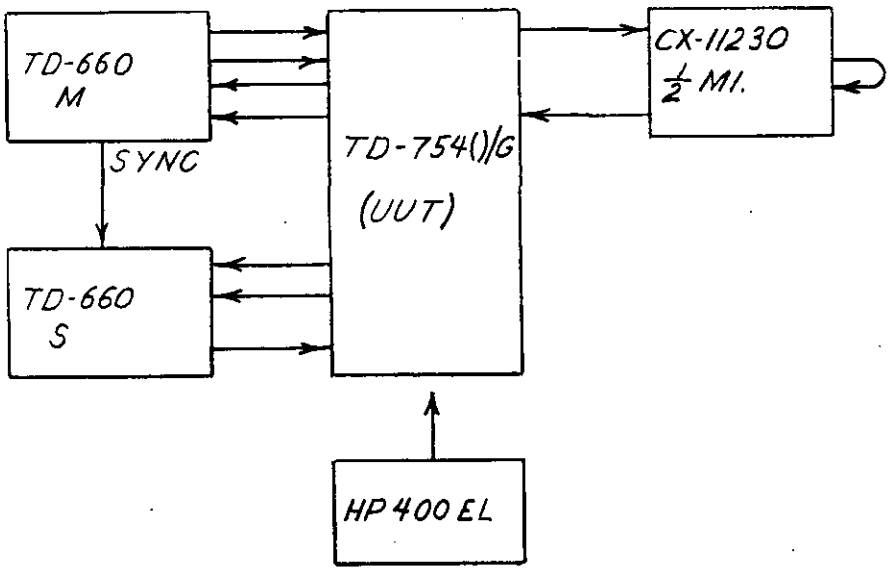
1. TRAFFIC ALARM
2. CABLE PWR. ALARM
3. LIMITED FAULT LOCATION

FIG. 3

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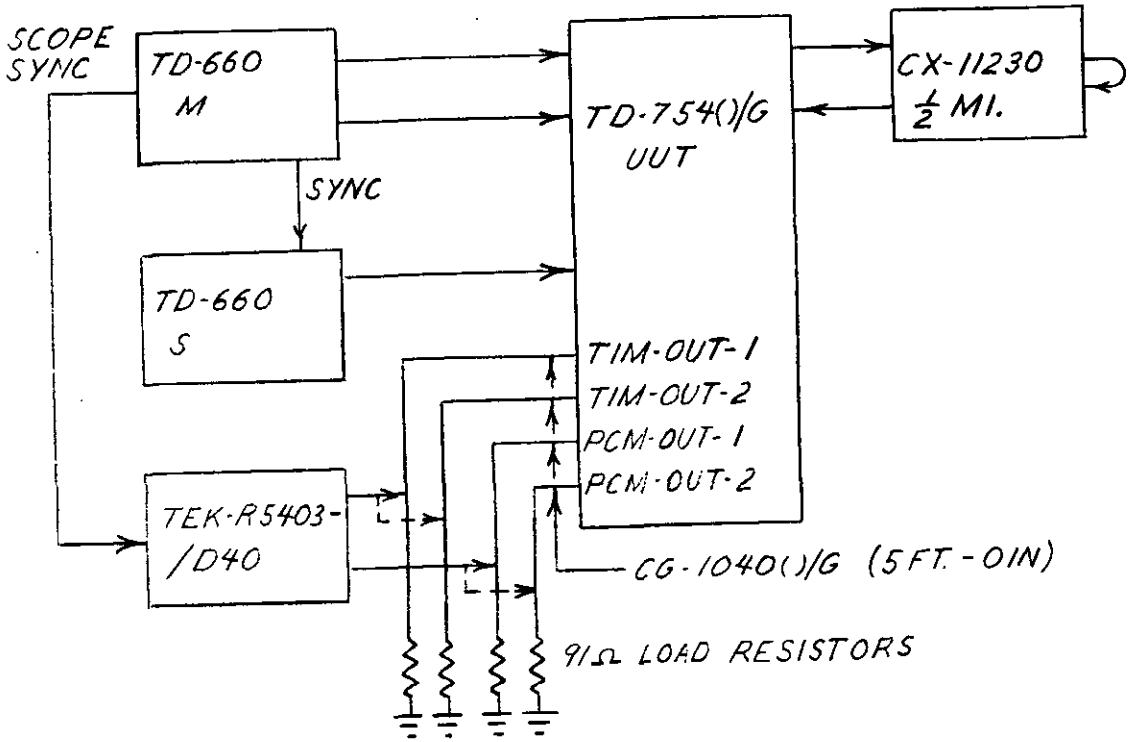


TEST TONE TEST
FIG. 4

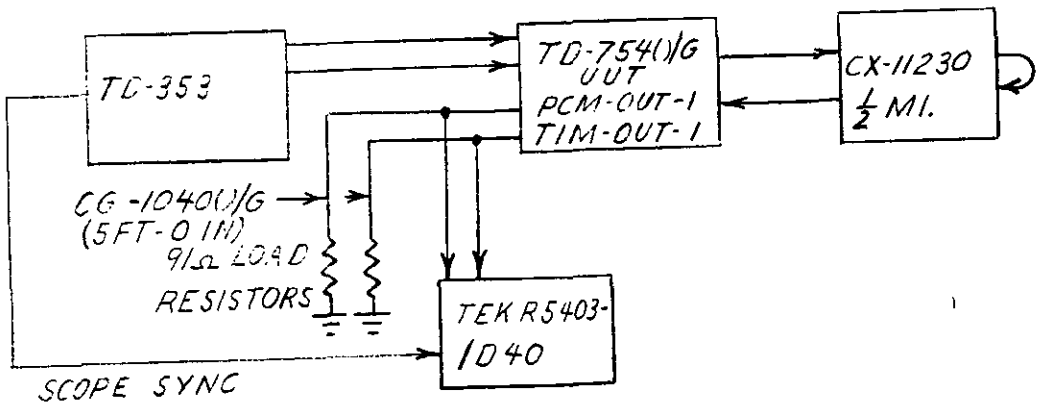


METERING TEST
FIG. 5

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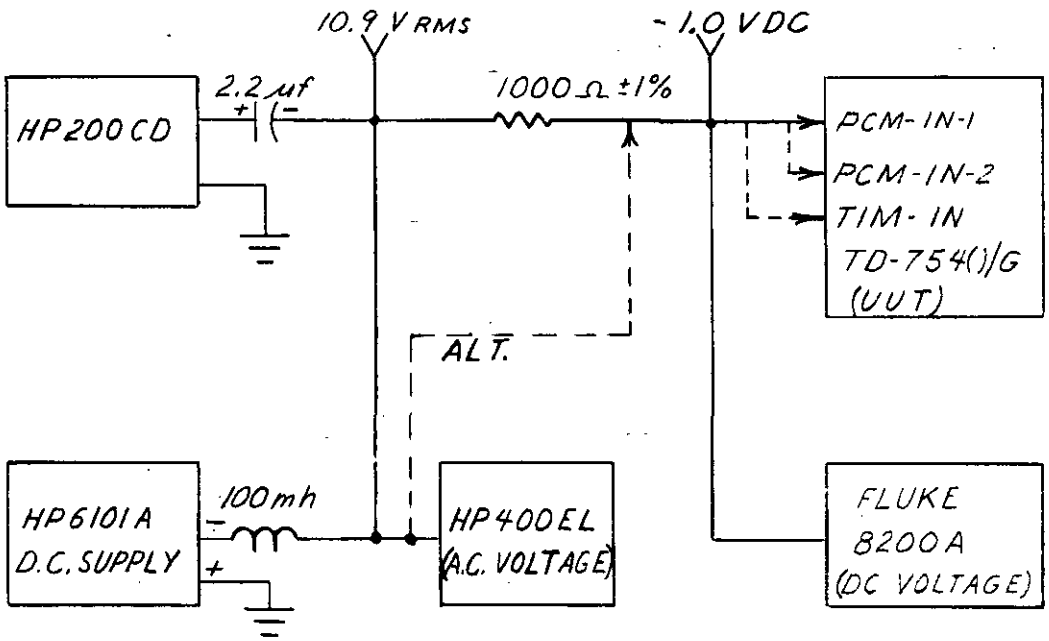


DIGITAL SIGNAL TEST
FIG. 6

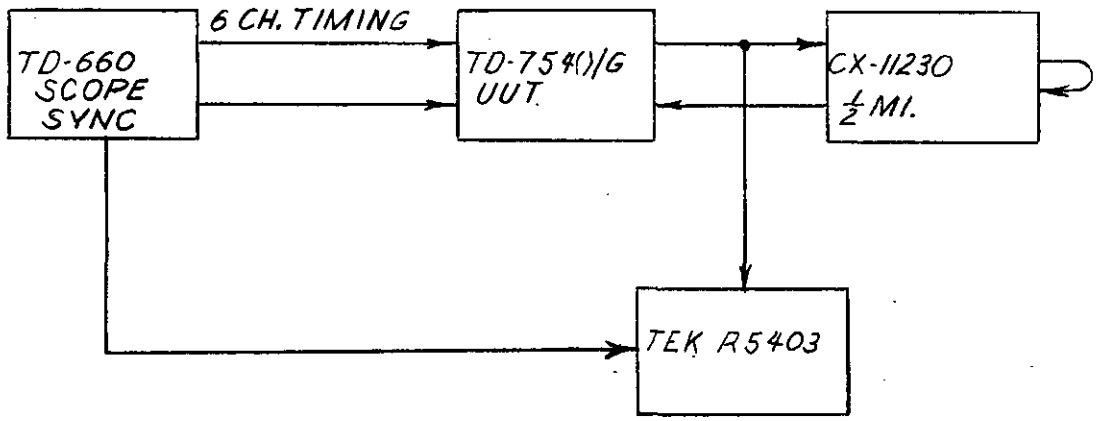


DIGITAL SIGNAL TEST
FIG. 7

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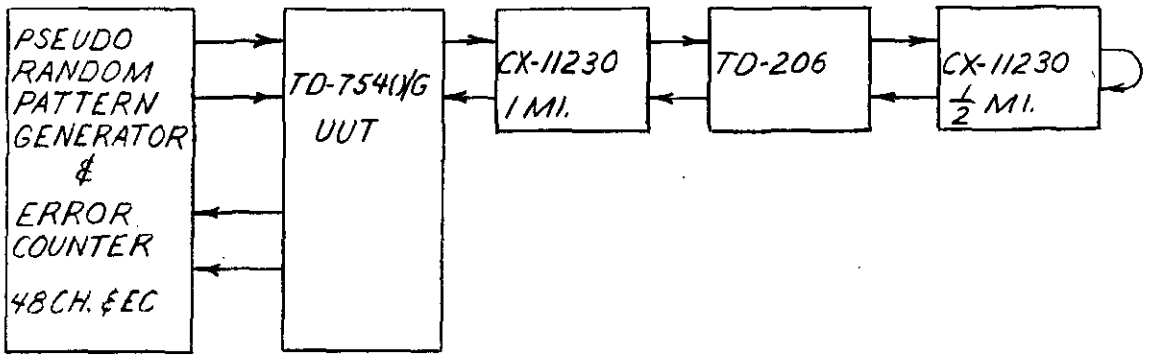


DIGITAL IMPEDANCE TEST
FIG. 8



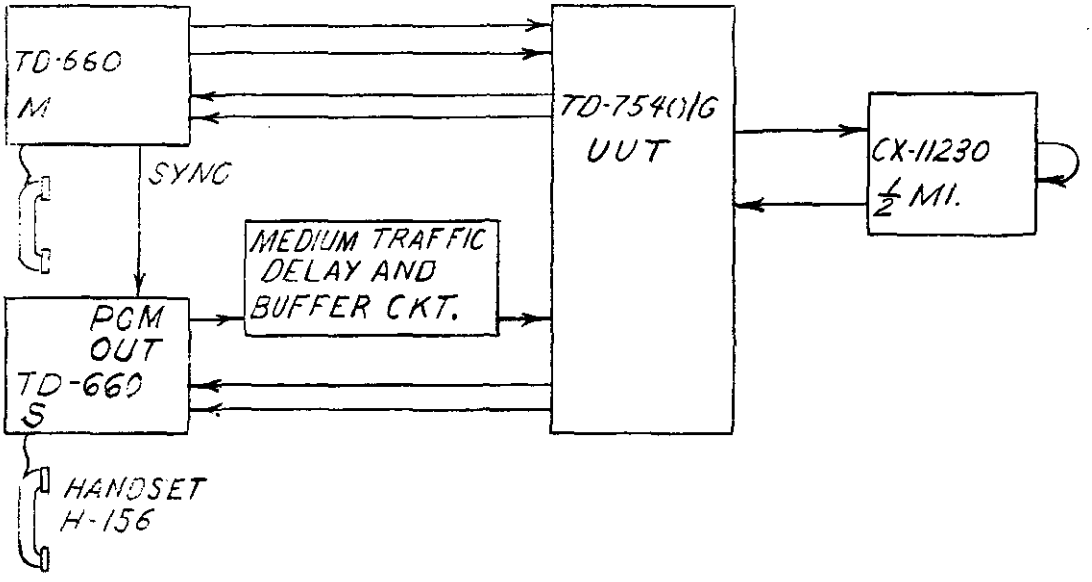
CABLE SIGNAL TEST
FIG. 9

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¼, ½, ¾, 1 MILE TEST

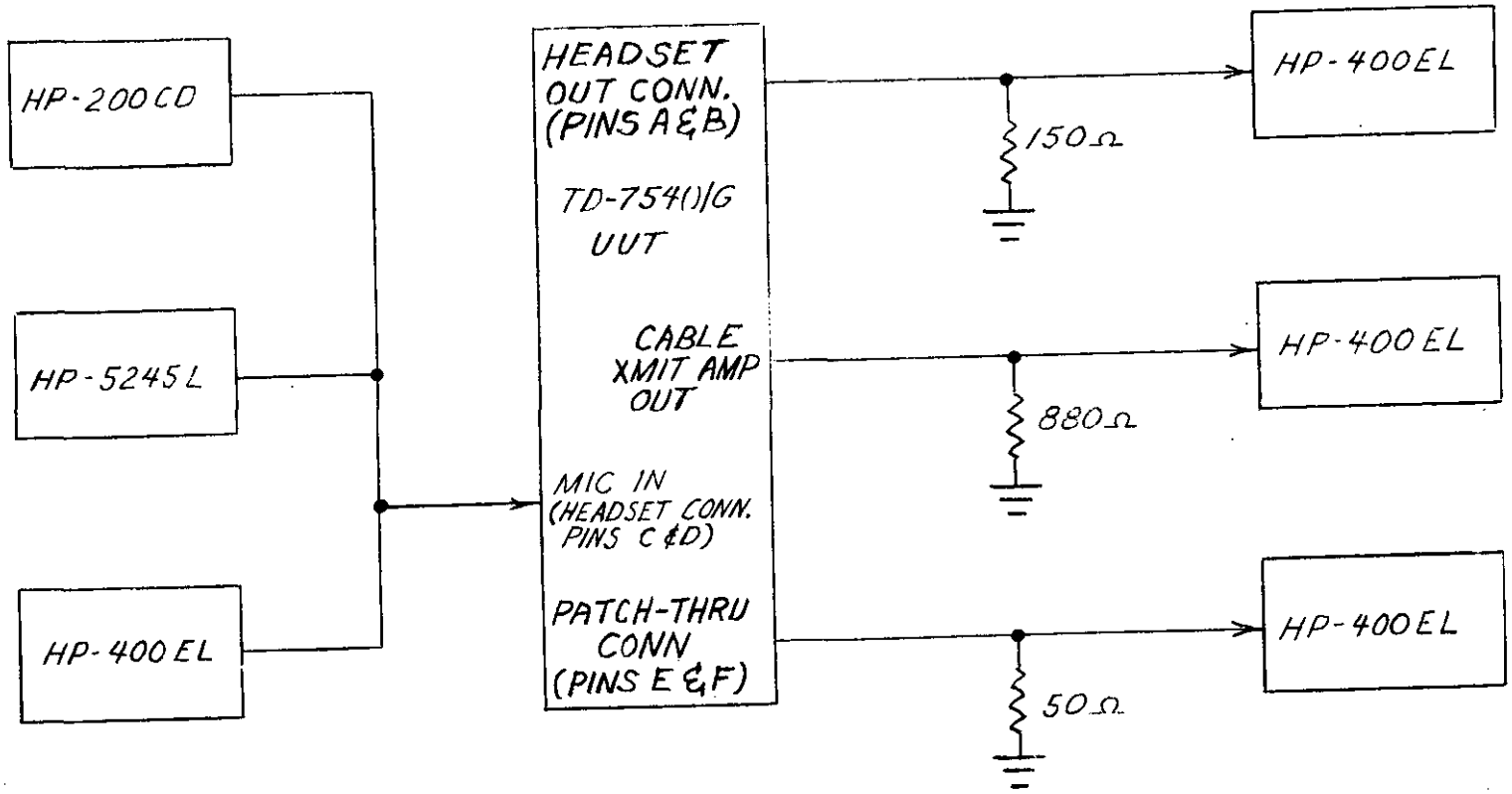
FIG. 10



AUTOMATIC PHASING TEST

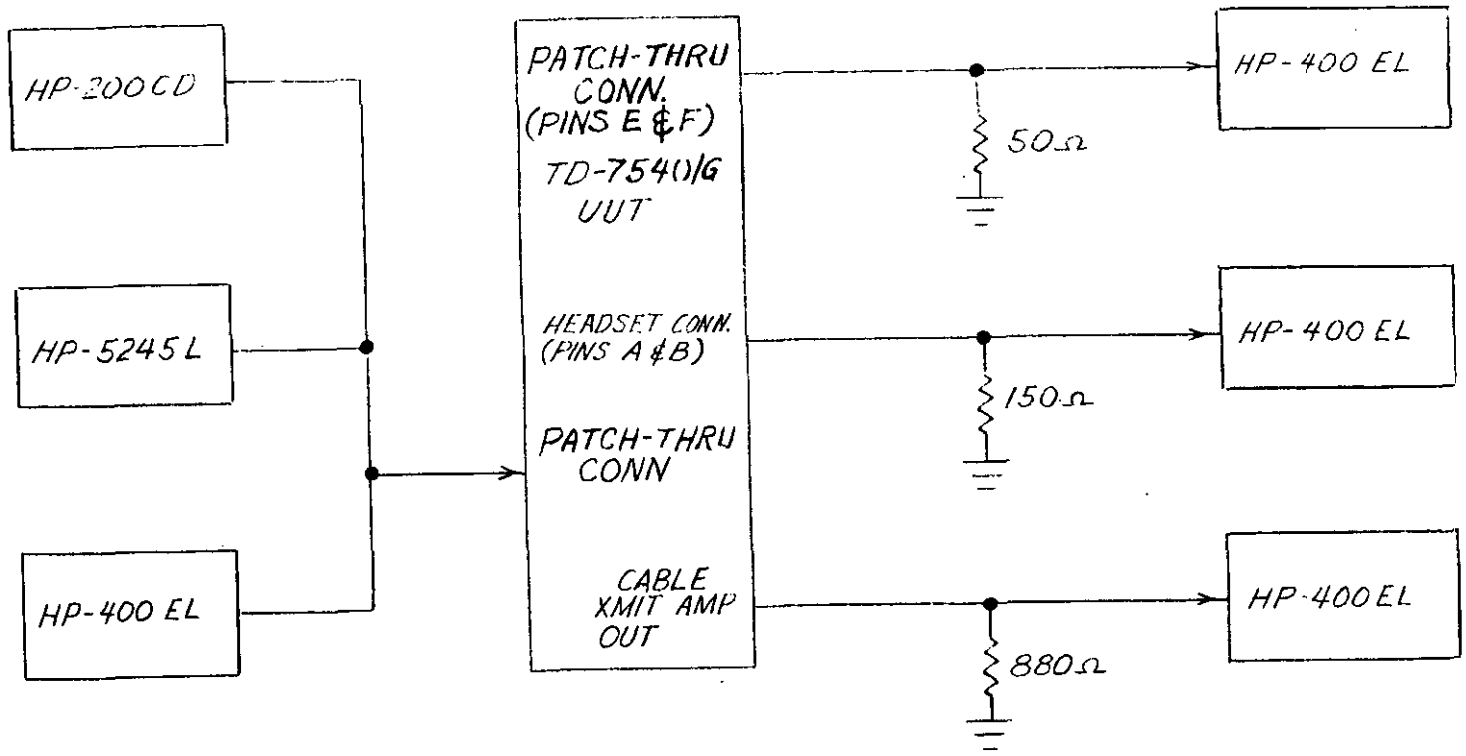
FIG. 11

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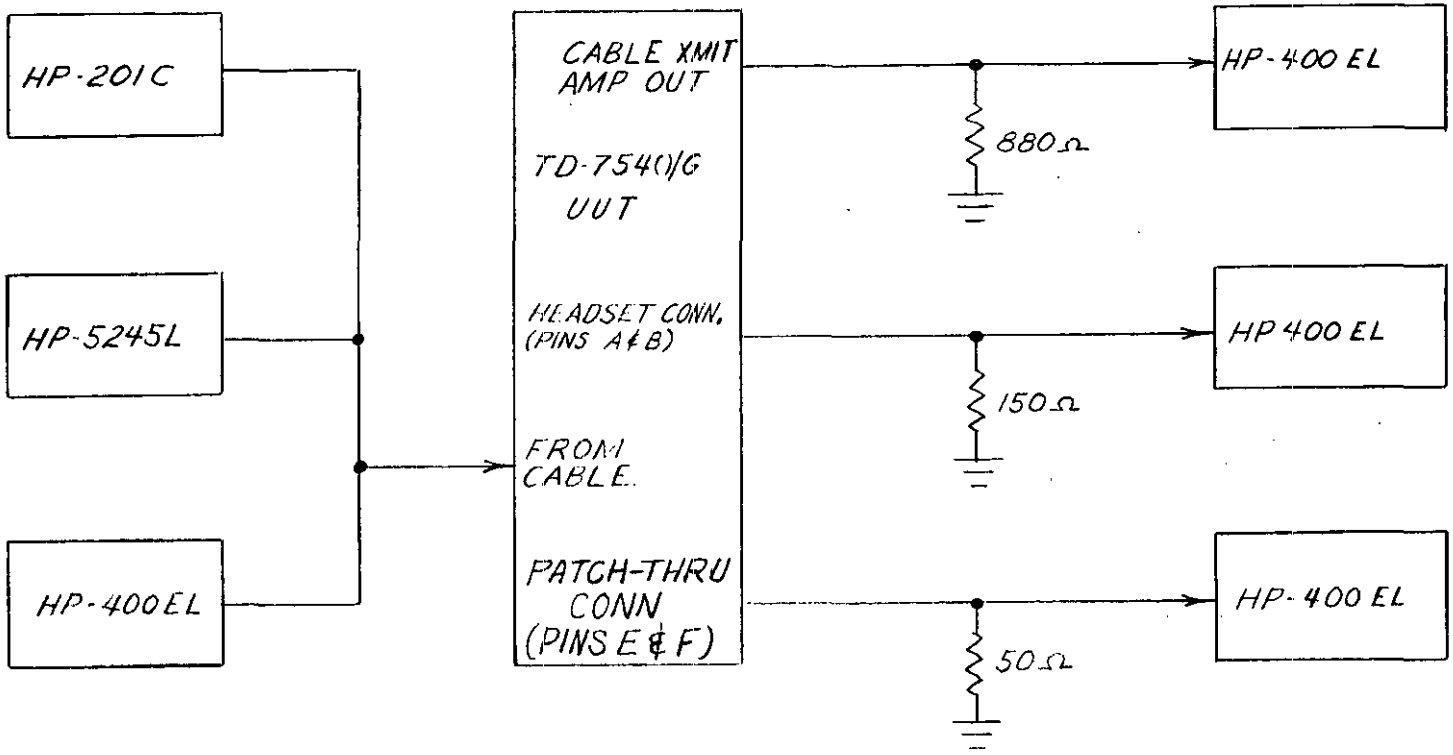
TRANSMIT AMP TEST
FIG. 12

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REC. AMP (R/P.T.) TEST
FIG. 13

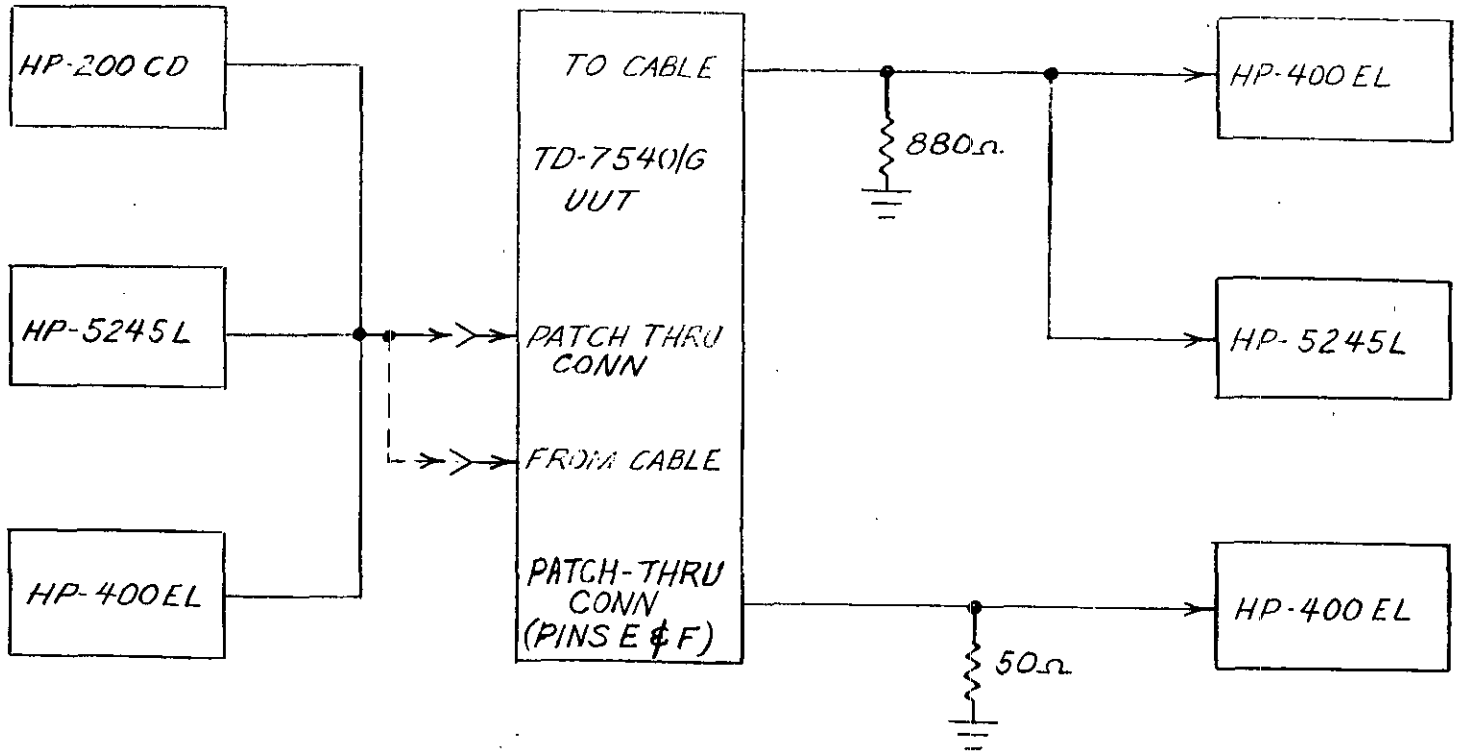
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RECEIVE AMP(CR) TEST

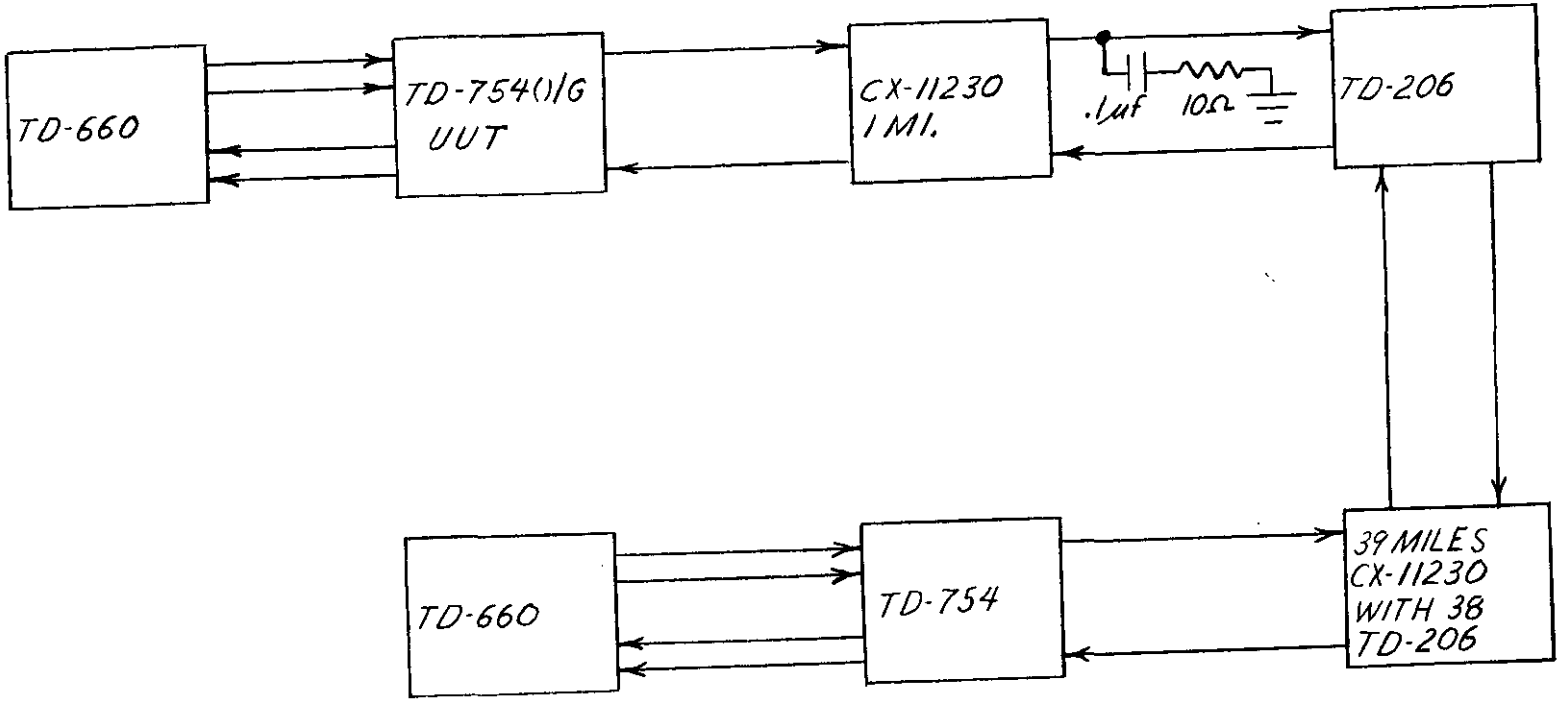
FIG. 14

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SIGNALING TEST
FIG. 15

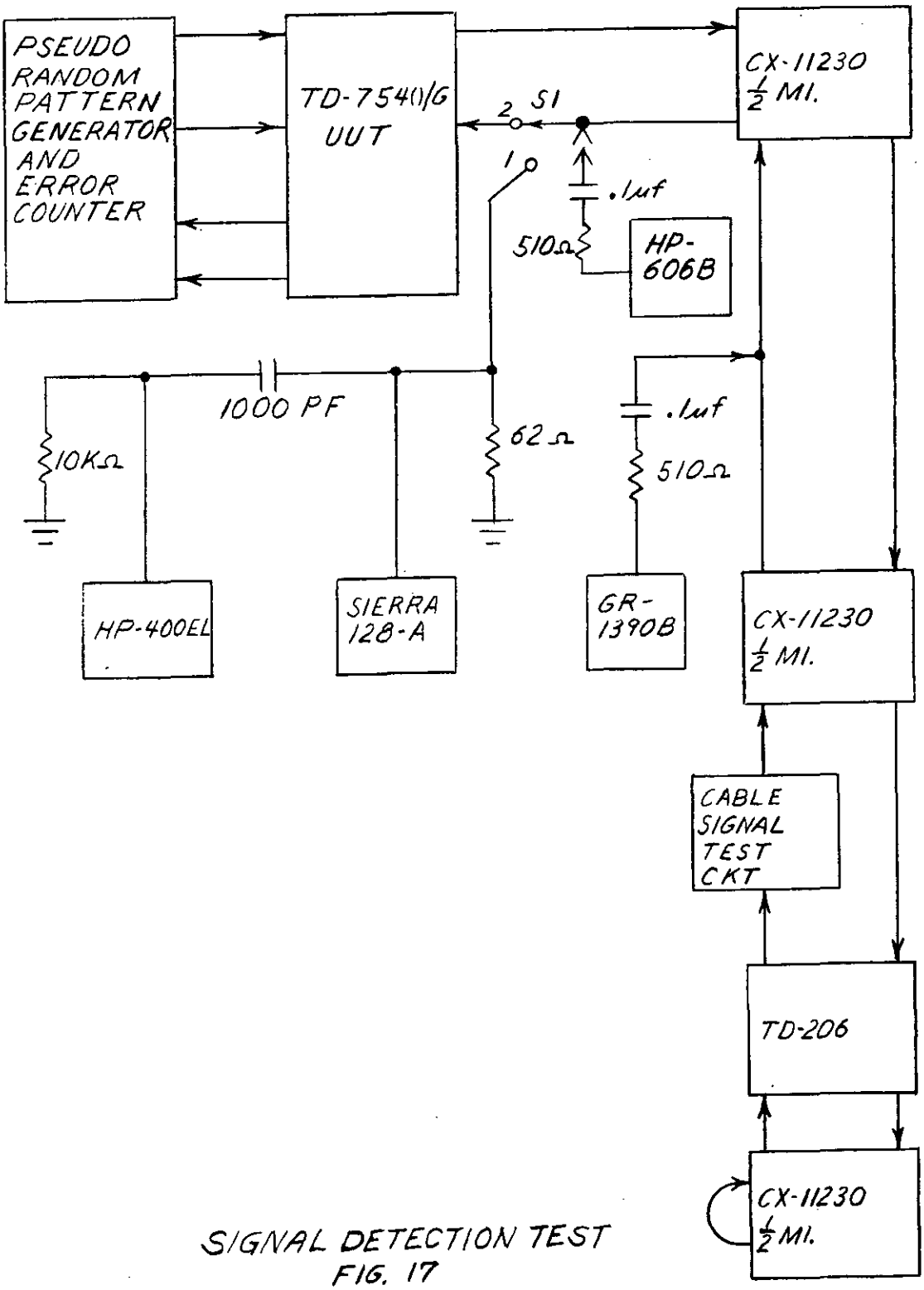
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FAULT LOCATION TEST

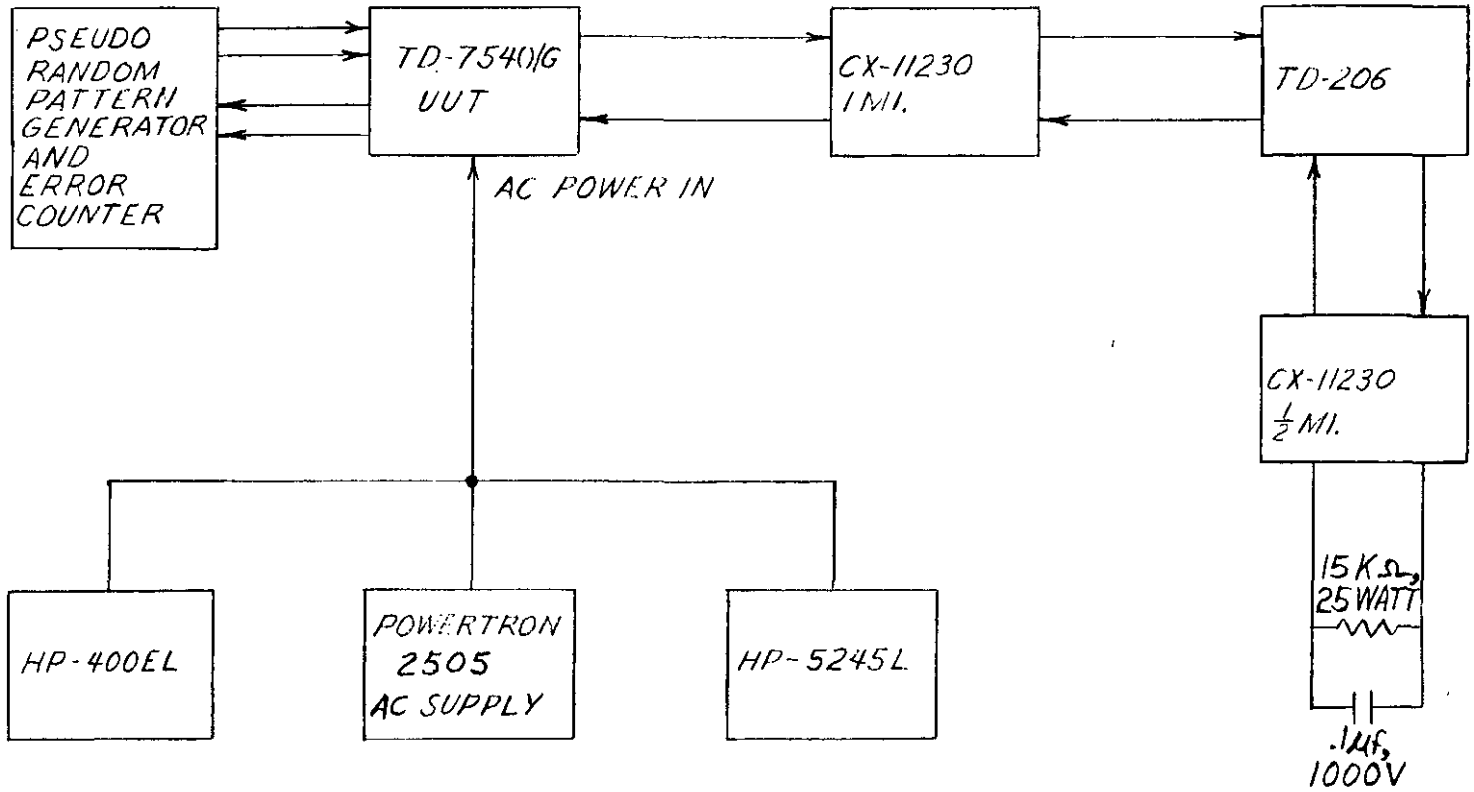
FIG. 16

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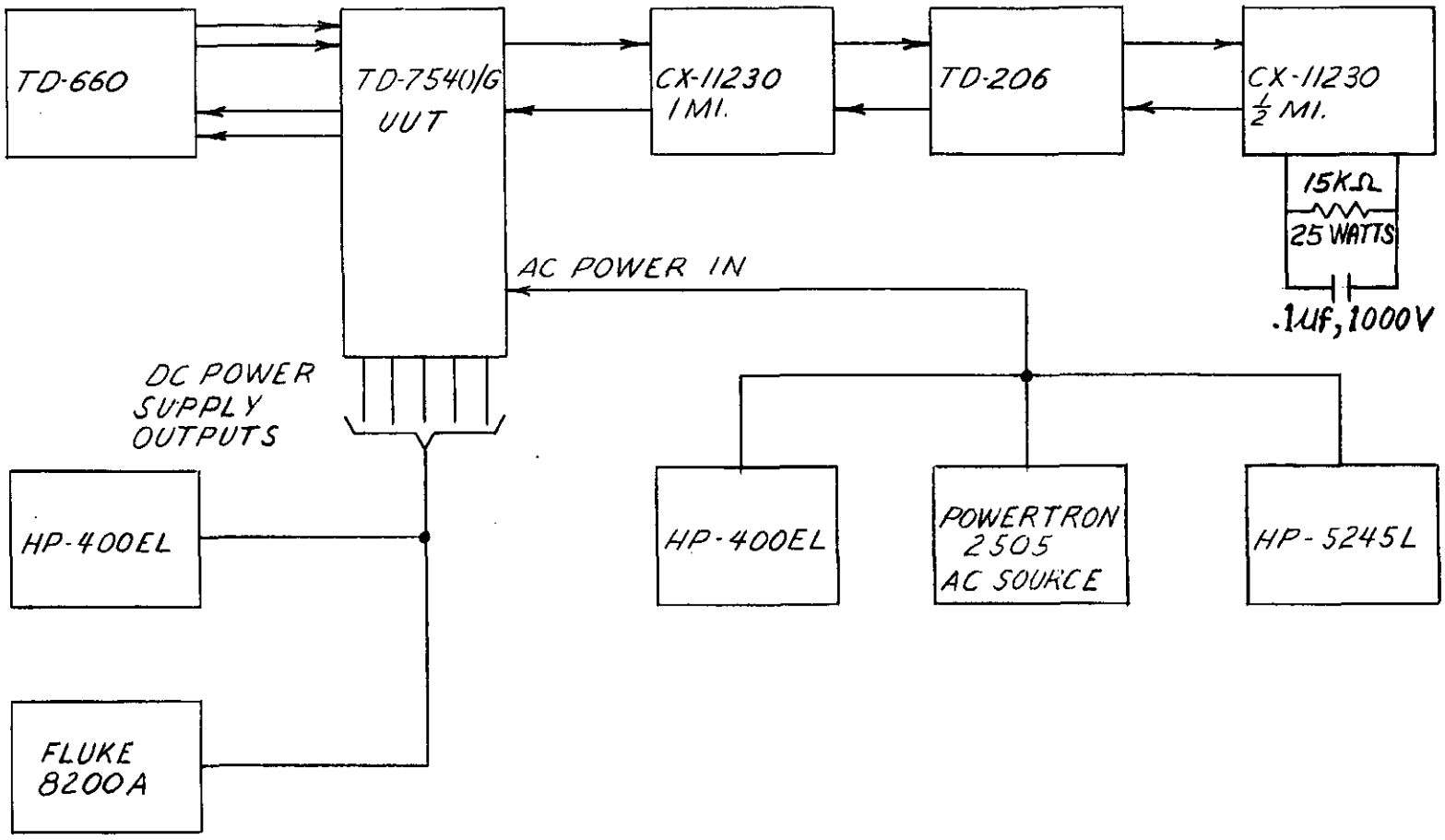
SIGNAL DETECTION TEST
FIG. 17

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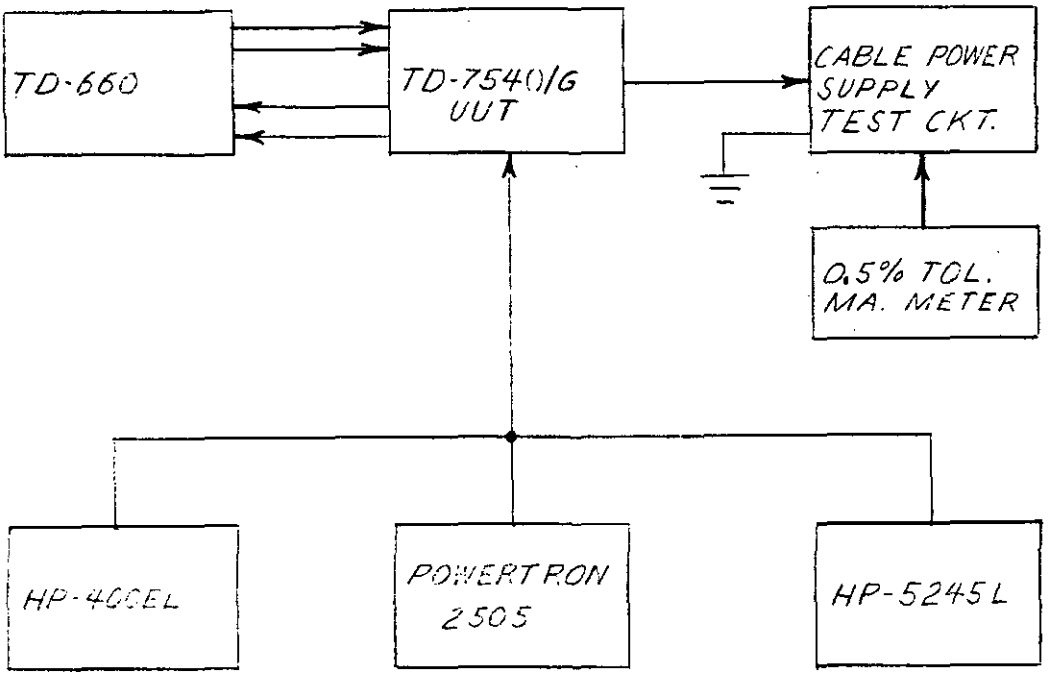
AC INPUT TEST
FIG. 18

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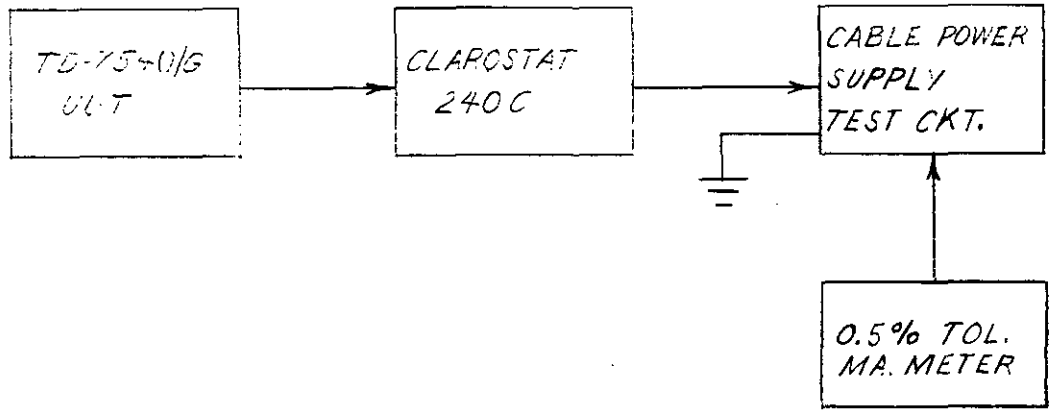


POWER SUPPLY OUTPUT TEST
FIG. 19

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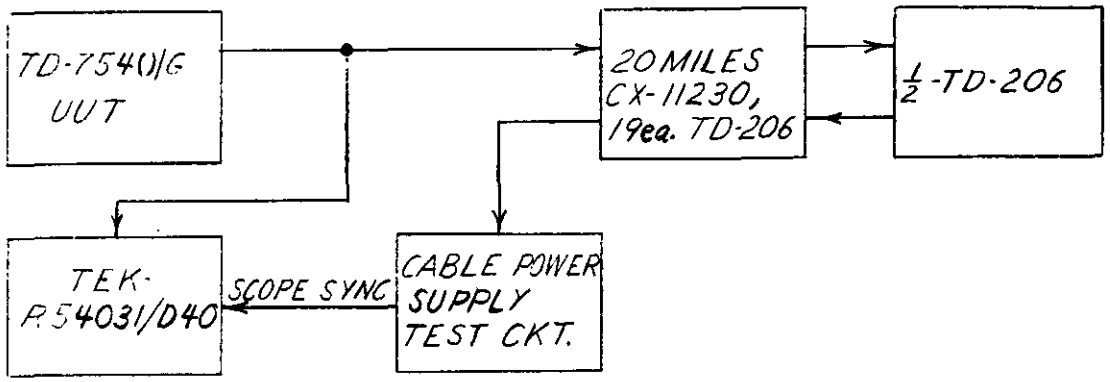


CABLE POWER SUPPLY TEST
FIG. 20



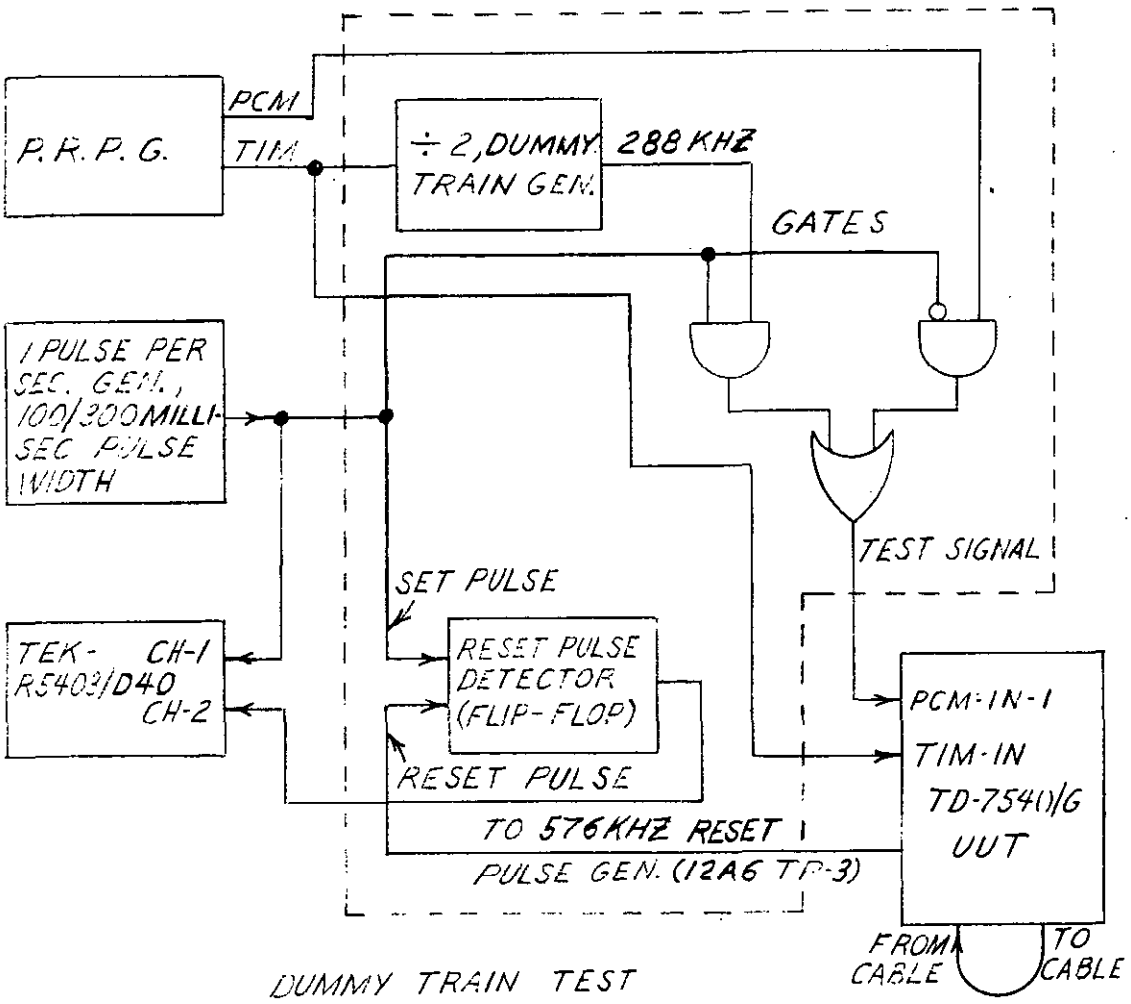
AUTOMATIC SHUTOFF TEST
FIG. 21

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AUTOMATIC SHUTOFF TEST

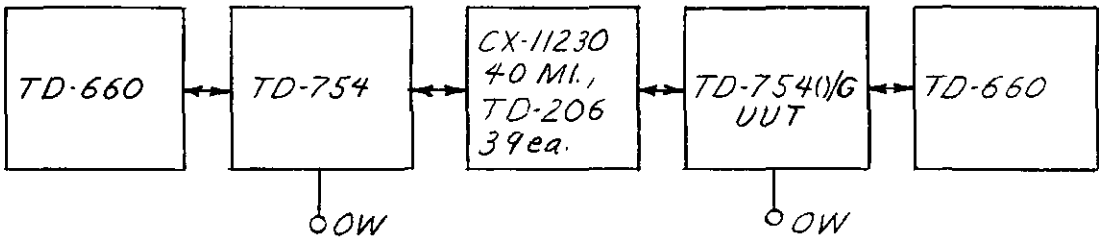
FIG. 22



DUMMY TRAIN TEST

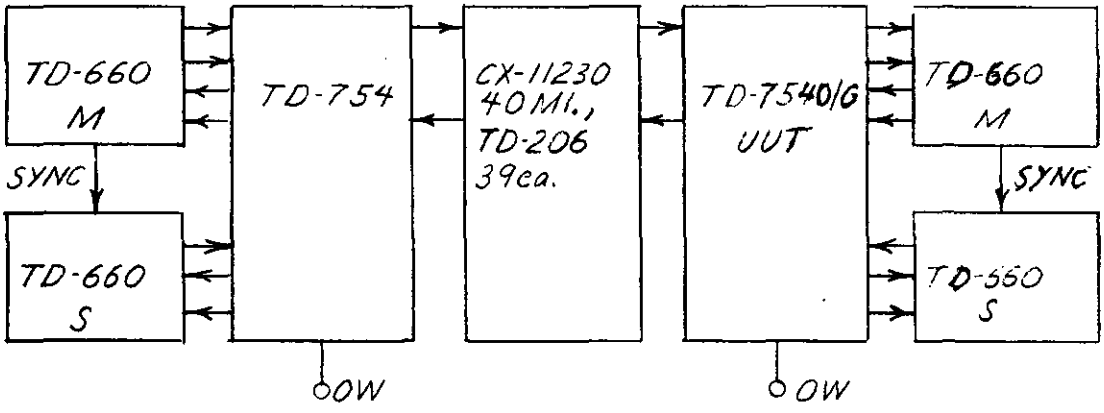
FIG. 23

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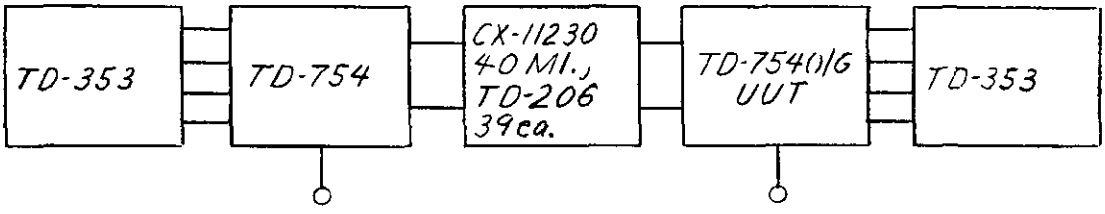
6/12 CHANNEL CABLE SYSTEM TEST

FIG. 24



24 CHANNEL CABLE SYSTEM TEST

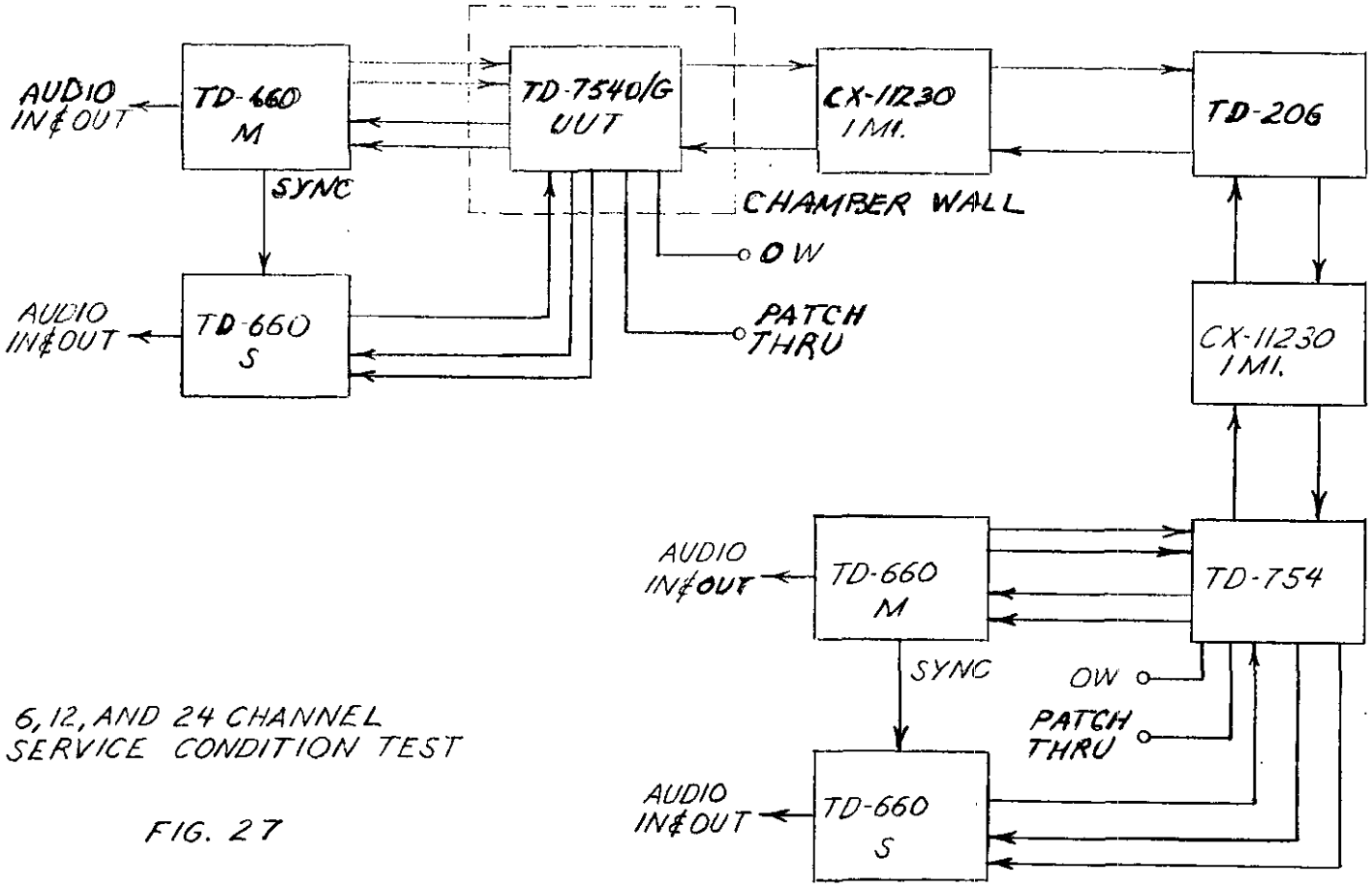
FIG. 25



48 CHANNEL CABLE SYSTEM TEST

FIG. 26

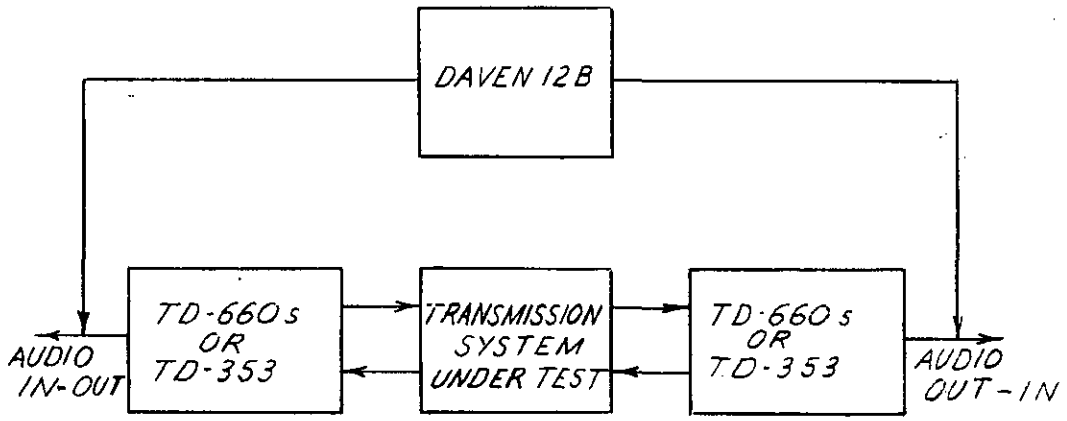
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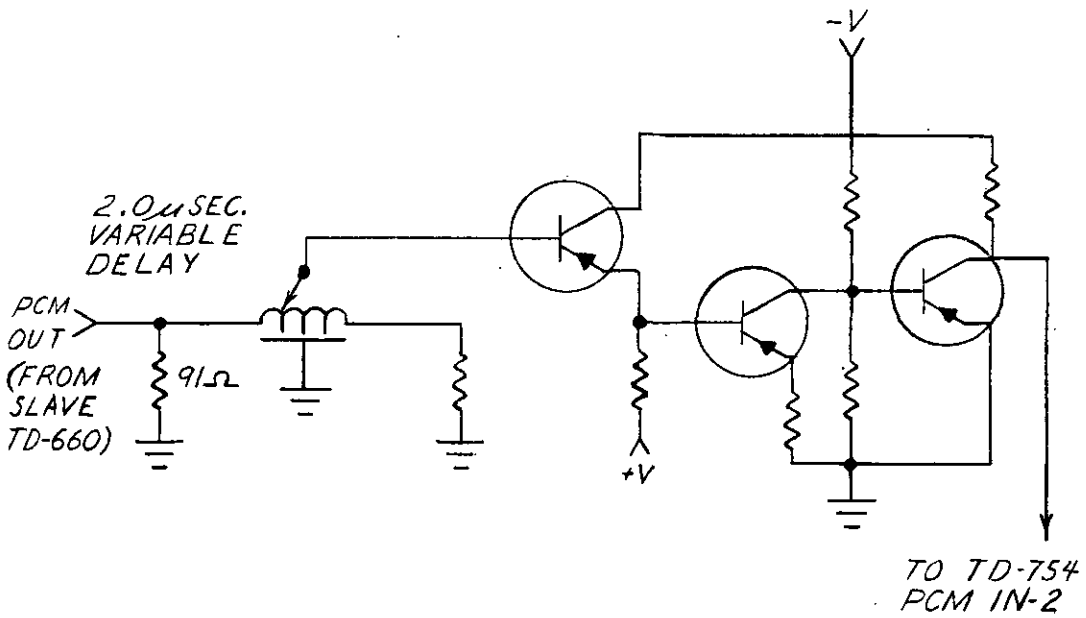
6, 12, AND 24 CHANNEL SERVICE CONDITION TEST

FIG. 27

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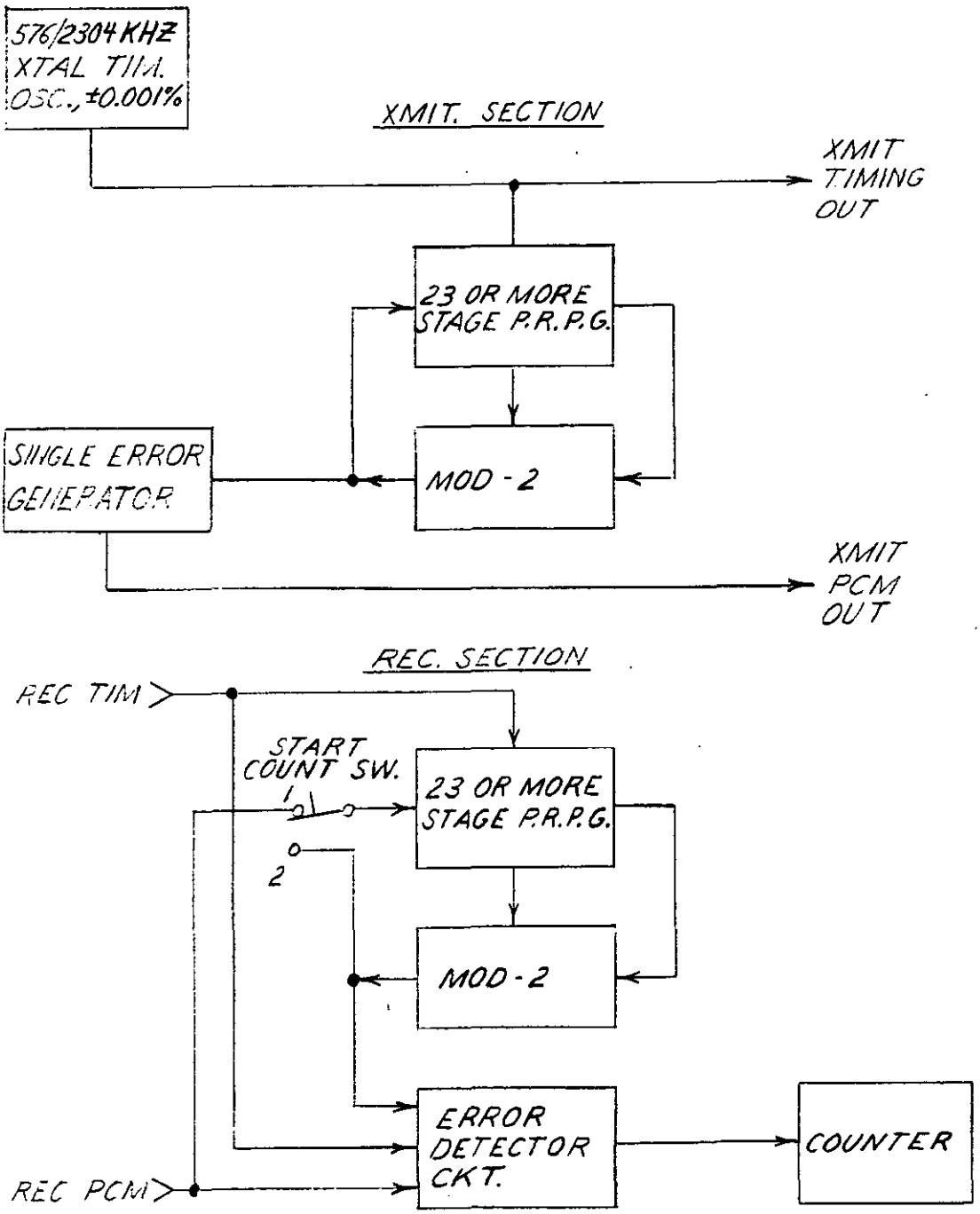


SIGNAL TO NOISE RATIO & SIGNAL TO DISTORTION
FIG. 28



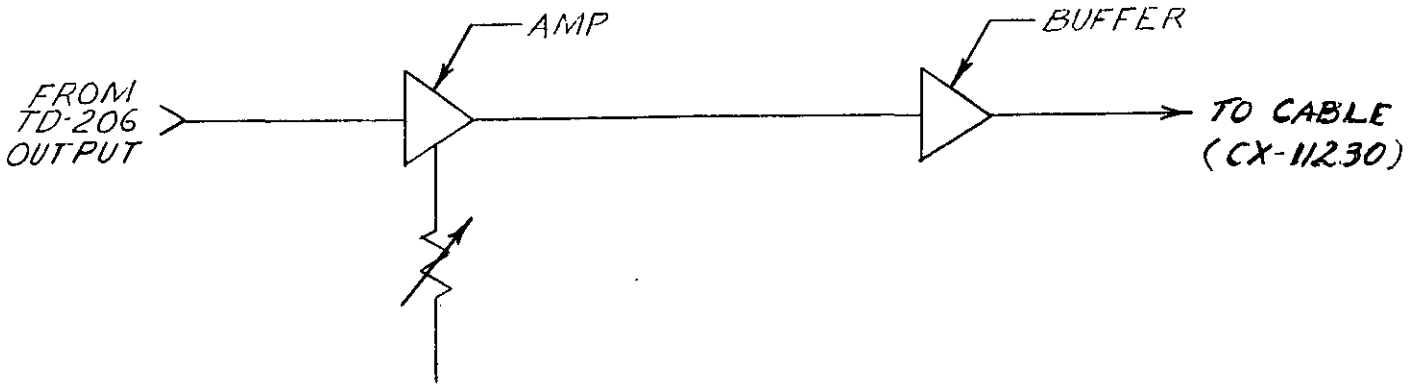
MEDIUM TRAFFIC DELAY & BUFFERING CKT.
FIG. 29

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PSEUDO RANDOM PATTERN GENERATOR (P. R. P. G.)
AND ERROR COUNTER
FIG. 30

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NOTES:

1. ALSO, DC & OW MUST PASS THRU.
2. AMP GAIN MUST BE ADJUSTABLE TO PROVIDE 20 TO 50 MV PEAK TO PEAK AT THE TD-754()/G INPUT THRU 1 MILE OF CX-11230 CABLE.

CABLE SIGNAL TEST CKT.

FIG. 31

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CABLE POWER SUPPLY TEST CKT.

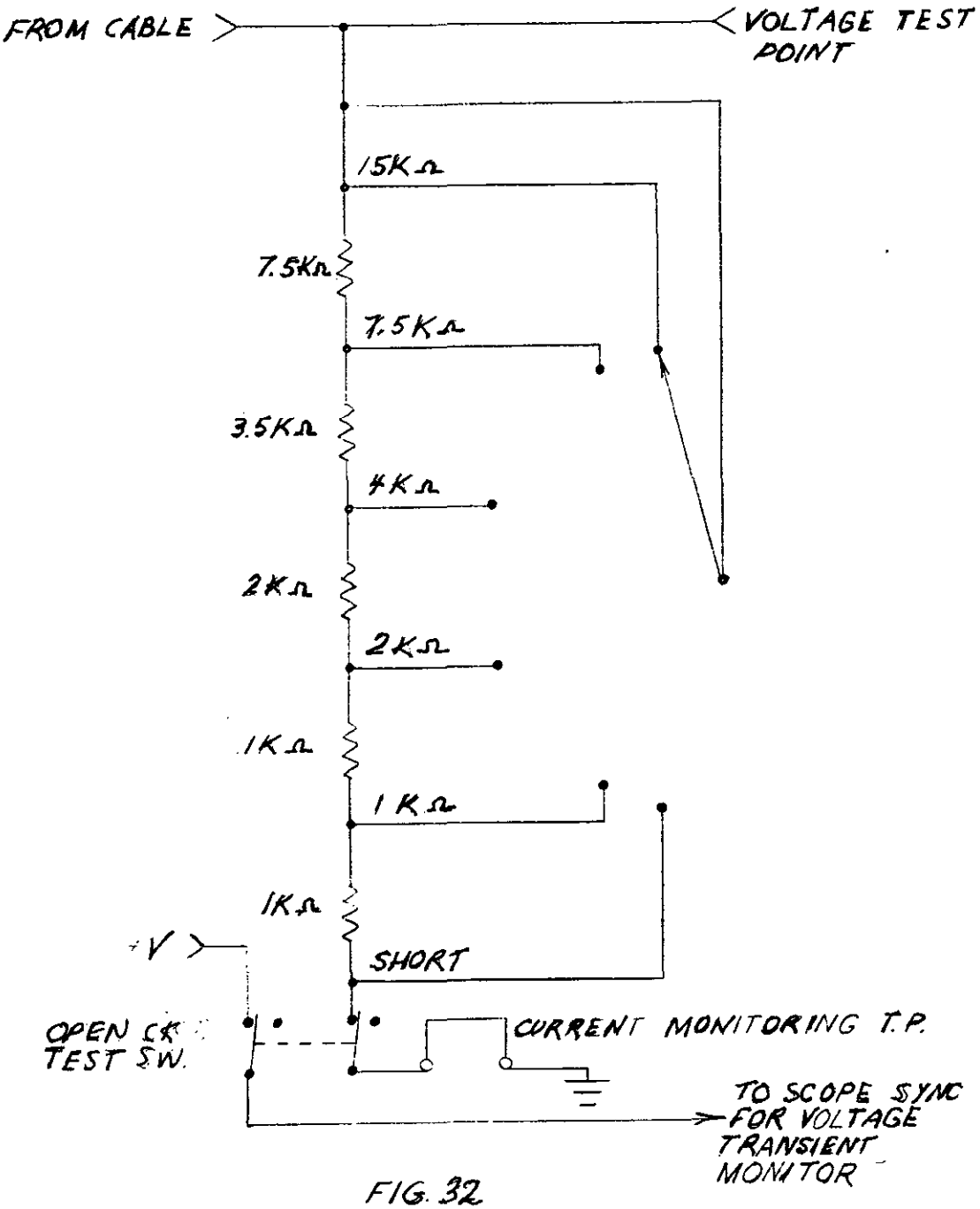
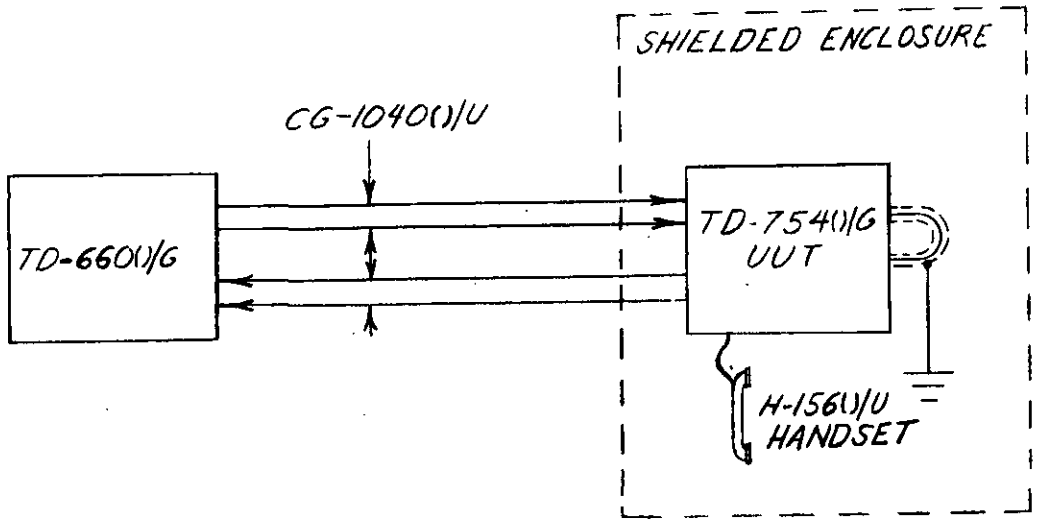


FIG. 32

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ELECTROMAGNETIC COMPATIBILITY TEST
FIG. 33

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-55649A(EL) MULTIPLEXER TD-754( )/G

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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